The Annals of Statistics
2026, Vol. 54, No. 3, 1262–1288
https://doi.org/10.1214/25-AOS2603

STATISTICAL-COMPUTATIONAL TRADE-OFFS FOR RECURSIVE ADAPTIVE PARTITIONING ESTIMATORS

BY YAN SHUO TAN[1,a], JASON M. KLUSOWSKI[2,b] AND KRISHNAKUMAR BALASUBRAMANIAN[3,c]

[1]*Department of Statistics and Data Science, National University of Singapore,* [a]*yanshuo@nus.edu.sg*
[2]*Department of Operations Research and Financial Engineering, Princeton University,* [b]*jason.klusowski@princeton.edu*
[3]*Department of Statistics, University of California, Davis,* [c]*kbala@ucdavis.edu*

Models based on recursive adaptive partitioning such as decision trees and their ensembles are popular for high-dimensional regression as they can potentially avoid the curse of dimensionality. Because empirical risk minimization (ERM) is computationally infeasible, these models are typically trained using greedy algorithms. Although effective in many cases, these algorithms have been empirically observed to get stuck at local optima. We explore this phenomenon in the context of learning sparse regression functions over d binary features, showing that when the true regression function f^* does not satisfy Abbe et al. (In *Conference on Learning Theory* (2022) 4782–4887 PMLR)'s Merged Staircase Property (MSP), a form of heredity restriction similar to that used in classical ANOVA modeling, greedy training requires $\exp(\Omega(d))$ samples to achieve low estimation error. Conversely, when f^* does satisfy MSP, greedy training can attain small estimation error with only $O(\log d)$ samples. This dichotomy mirrors that of two-layer neural networks trained with stochastic gradient descent (SGD) in the mean-field regime, thereby establishing a head-to-head comparison between SGD-trained neural networks and greedy recursive partitioning estimators. Furthermore, ERM-trained recursive partitioning estimators achieve low estimation error with $O(\log d)$ samples irrespective of whether f^* satisfies MSP, thereby demonstrating a statistical-computational trade-off for greedy training. Our proofs are based on a novel interpretation of greedy recursive partitioning using stochastic process theory and a coupling technique that may be of independent interest.

1. Introduction. Decision tree models are piecewise constant supervised learning models obtained by recursive adaptive partitioning of the covariate space. Although classical, they remain among the most important supervised learning models because they are highly interpretable [40, 43] and yet are flexible enough to afford the potential for high prediction accuracy. This potential is maximized when decision trees are combined in ensembles via random forests (RFs) [12] or gradient boosting [23]. These algorithms are widely recognized as having state-of-the-art performance on moderately-sized tabular datasets [14, 22, 41], even outperforming state-of-the-art deep learning methods [25], despite the amount of attention lavished on the latter. Such datasets are common in many settings such as bioinformatics, healthcare, economics and social sciences. Naturally, decision trees and their ensembles receive widespread use via their implementation in popular machine learning packages such as `ranger` [56], `scikit-learn` [42], `xgboost` [17] and `lgbm` [30]. Decision trees have also been adapted to a variety of tasks beyond regression and classification, including survival analysis [28], heterogeneous treatment effect estimation [9], time-series analysis and multitask learning.

Received November 2024; revised November 2025.
MSC2020 subject classifications. 62G08, 68Q32.
Key words and phrases. CART, random forest, greedy, tree, recursive partitioning, statistical-computational trade-offs.

While alternatives exist, most decision tree models used in practice make binary, axis-aligned splits at each partitioning stage. In this paper, we study the *statistical-computational trade-offs* of these objects, focusing on regression trees. We assume a nonparametric regression model under random design:

$$Y_i = f^*(\mathbf{X}_i) + \varepsilon_i, \quad i = 1, 2, \dots, n. \tag{1}$$

Here, the covariate space $\mathcal{X}$ is a compact subset of $\mathbb{R}^d$, $\mathbf{X}_1, \mathbf{X}_2, \dots \mathbf{X}_n$ are drawn i.i.d. from a distribution ν on $\mathcal{X}$, $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ are noise variables drawn i.i.d. from a zero-mean noise distribution on $\mathbb{R}$, with $\varepsilon_i \perp\!\!\!\perp \mathbf{X}_i =: (X_{i1}, X_{i2}, \dots, X_{id})$, and $f^*\colon \mathcal{X} \to \mathbb{R}$ is the conditional expectation function of the response Y given $\mathbf{X} = \mathbf{x}$. The observed data is denoted as $\mathcal{D}_n := \{(\mathbf{X}_i, Y_i)\colon i = 1, \dots, n\}$. We will compare and contrast various tree-based estimators for f^*, evaluating the accuracy of an estimate $\hat{f}(-; \mathcal{D}_n)$ using the L^2 risk (or estimation error)

$$R(\hat{f}(-; \mathcal{D}_n), f^*) := \mathbb{E}_{\mathbf{X}\sim\nu}\{(\hat{f}(\mathbf{X}; \mathcal{D}_n) - f^*(\mathbf{X}))^2\},$$

and the accuracy of an estimator $\hat{f}(-; -)$ using the expected L^2 risk, $\mathbb{E}_{\mathcal{D}_n}\{R(\hat{f}(-; \mathcal{D}_n), f^*)\}$. Note that throughout this paper, we will use the convention that random variables are denoted in upper case while any fixed value they may attain is denoted in lower case. Furthermore, vectors will be denoted using boldface, whereas scalars will be denoted using regular font. We denote $[d] := \{1, 2, \dots, d\}$. To supplement standard Big-O notation, we use $\tilde{O}$ to suppress poly-logarithmic factors.

1.1. *High-dimensional consistency.* To establish statistical-computational trade-offs for regression trees, we analyze their performance in a high-dimensional setting ($n, d \to \infty$) with sparsity constraints on f^*, that is, if f^* only depends on a constant number of s covariates. Informally, we say that an estimator avoids the curse of dimensionality, or that it is *high-dimensional consistent*, if its expected L^2 risk converges to zero under a scaling regime satisfying $\log n = o(d)$. It is instructive to note that high-dimensional consistency necessarily requires some form of feature selection. Under relatively balanced splits, any leaf containing a constant number of data points has depth $O(\log n)$, so along a root-to-leaf path at most $O(\log n)$ distinct covariates can be used for splitting. When $\log n = o(d)$, this means that most coordinates are never split along that path, so the regression function cannot be resolved in those directions. Consequently, totally randomized trees, as well as other procedures that grow decision trees independently of the observed responses, cannot be high-dimensional consistent (see also Section B of the Supplementary Material [47]).

Several works have showed that CART and RFs are high-dimensional consistent given various condition on f^* (e.g., [45] for binary covariates and assuming submodularity, [33] for arbitrary covariates and assuming additivity, [18] for arbitrary covariates and assuming a condition called sufficient impurity decrease, which we will define more formally later in this paper). This helps to explain the successful application of RFs to high-dimensional problems such as genomics in which $n \ll d$.

On the other hand, these works were not able to show whether their assumed conditions were *necessary* for high-dimensional consistency. [45] used the exclusive-or (XOR) function to argue heuristically that some condition is indeed necessary (see Section 5 therein). To understand this argument, we focus on the setting of binary covariates ($\mathcal{X} = \{\pm 1\}^d$) under the uniform distribution. In this setting, the XOR function can be written as $f^*(\mathbf{x}) = x_1 x_2$. Meanwhile, every tree node corresponds to a subcube $\mathcal{C} \subset \{\pm 1\}^d$ and every potential split of $\mathcal{C}$ divides it evenly into two subcubes of lower dimension. With each split fully determined by the choice of covariate index $k = 1, \dots, d$, CART chooses the index of the covariate with the maximum plug-in squared correlation with the response within $\mathcal{C}$:

$$\widehat{\mathrm{Corr}}^2\{Y, X_k \mid \mathbf{X} \in \mathcal{C}\} = \frac{(\sum_{\mathbf{X}_i \in \mathcal{C}} X_{ik}(Y_i - \bar{Y}_{\mathcal{C}}))^2}{\sum_{\mathbf{X}_i \in \mathcal{C}} (X_{ik} - (\bar{X}_k)_{\mathcal{C}})^2 \sum_{\mathbf{X}_i \in \mathcal{C}} (Y_i - \bar{Y}_{\mathcal{C}})^2}; \tag{2}$$

(see [32] and [31], equation (5)). Unless $\mathcal{C}$ already depends on either x_1 or x_2, however, we have zero correlation in the infinite-sample limit, that is,

$$\operatorname{Corr}^2\{f^*(\mathbf{X}), X_k \mid \mathbf{X} \in \mathcal{C}\} = \mathbb{E}\{X_1 X_2 X_k \mid \mathbf{X} \in \mathcal{C}\}^2 = 0, \tag{3}$$

for $k = 1, \dots, d$. This implies that CART cannot differentiate between relevant and irrelevant covariates, instead tending to make nonadaptive random splits. Hence, it cannot avoid the curse of dimensionality. As an ensemble of CART trees, RF suffers from the same limitations.

Beyond some heuristic calculations, [45] did not make this argument rigorous, nor did they generalize beyond the XOR example.

> *In this paper, we seek to bridge this gap and establish a necessary and nearly sufficient condition on f^* that characterizes when CART is high-dimensional consistent, and, in doing so, explicitly compare its risk performance against that of ERM and two-layer neural networks trained with SGD.*

We do this in the context of binary covariates under the uniform distribution, which although unlikely to describe any real dataset, provides a useful setting where both results and calculations can be stated cleanly and precisely, thereby illustrating the essence of the strengths and limitations of using a greedy splitting rule.

1.2. *"Almost" characterization using the merged-staircase property.* For any subset $S \subset [d]$ of the coordinate indices, let χ_S be the monomial function defined by $\chi_S(\mathbf{x}) = \prod_{j \in S} x_j$.[1] It is a classical result from Fourier analysis [44] that every function $f^* \colon \{\pm 1\}^d \to \mathbb{R}$ can be uniquely written in the form

$$f^* = \sum_{i=1}^{r} \alpha_{S_i} \chi_{S_i}, \tag{4}$$

for some nonnegative integer r, subsets $S_1, S_2, \dots, S_r$ of $\{1, 2, \dots, d\}$ and nonzero coefficients α_{S_j}, $j = 1, 2, \dots, r$. We say that f^* satisfies the *Merged-Staircase Property* (MSP) [1] if the subsets $S_1, S_2, \dots, S_r$ can be reordered such that for any $i \in \{1, 2, \dots, r\}$, we have

$$\left| S_i \setminus \bigcup_{j=1}^{i-1} S_j \right| \le 1. \tag{5}$$ [2]

Note that any monomial of degree $k > 1$ does not satisfy the MSP. In particular, the XOR function does not satisfy the MSP. On the other hand, a positive example satisfying MSP is provided by the function

$$f^*(\mathbf{x}) = x_1 + x_2 + x_1 x_2 x_3. \tag{6}$$

For further intuition, note that (4) can be interpreted as an ANOVA decomposition for $f^*(\mathbf{X})$ [53]. Each term χ_{S_i} is a contrast that captures either a main effect (if S_i is a singleton) or a k-way interaction (if S_i has cardinality $k > 1$). Meanwhile, the coefficient α_{S_i} represent the effect size. The MSP imposes a type of heredity[3] constraint on the pattern of interactions present in the ANOVA model: We can order the nonzero effects so that each newly-introduced effect brings in at most one new variable that has not appeared in any previously included

[1] If $|S| = s$, such a function is sometimes called an s-parity in the theoretical computer science literature.

[2] The merged-staircase property was first introduced by [1]. See also Section 1.6.

[3] The MSP is related but not equivalent to the conditions of weak and strong heredity often enforced by ANOVA modeling in practice [57]. Weak heredity, which asserts that an interaction is present only if at least one of its lower-order parents is present, is equivalent to the Staircase Property formulated by [4] to analyze neural networks, and is therefore stronger than the MSP. [4] later generalized the definition to the MSP, which turns out to offer a better characterization of learnability.

effect. In other words, interactions cannot appear "out of nowhere" combining many fresh variables at once.

Our main theorem, stated informally, is the following.

THEOREM 1.1 (Informal). *When* $\mathcal{X} = \{\pm 1\}^d$, $\nu = \mathrm{Unif}(\{\pm 1\}^d)$ *and* f^* *depends only on* s *covariates, then the expected* L^2 *risk of the CART estimator* $\hat{f}_{\mathrm{CART}}$ *satisfies*:

- (*Necessity*) *If* f^* *does not satisfy* MSP, *then* $\mathbb{E}_{\mathcal{D}_n}\{R(\hat{f}_{\mathrm{CART}}(-;\mathcal{D}_n), f^*)\} = \Omega(1)$ *whenever* $n = \exp(O(d))$.
- (*Near sufficiency*) *If* f^* *satisfies* MSP *and its Fourier coefficients* $\{\alpha_{S_i}\}_{i=1}^r$ *are generic*, *then* $\mathbb{E}_{\mathcal{D}_n}\{R(\hat{f}_{\mathrm{CART}}(-;\mathcal{D}_n), f^*)\} = O(2^s/n)$ *whenever* $n = \Omega(2^s \log d)$.

Furthermore, *regardless of whether* f^* *satisfies* MSP, *the expected* L^2 *risk of the ERM tree estimator satisfies* $\mathbb{E}_{\mathcal{D}_n}\{R(\hat{f}_{\mathrm{ERM}}(-;\mathcal{D}_n), f^*)\} = O(2^s \log d/n)$.

The formal statements of these results are provided in the forthcoming Theorems 5.1, 7.5 and 8.1, together with Propositions 6.1 and 6.2. Our upper bound on the risk is attained when using a minimum impurity decrease stopping condition (see (14) for details) with a fixed threshold γ that is chosen above the noise level but below the signal strength. The lower bound on the risk does not depend on γ or on the particular stopping/pruning choices.

Here, genericity of $\{\alpha_{S_i}\}_{i=1}^r$ holds with probability one whenever they are drawn from a distribution with a density (see the discussion in Section 7.1). Up to this asterisk, we see that high-dimensional consistency for CART can be completely characterized by MSP, a combinatorial condition on Fourier coefficients. In addition, we see in fact that CART behaves similar to ERM trees when MSP holds, and similarl to nonadaptive trees when it does not, thereby also establishing *performance gaps*: In the former case, CART sharply improves upon non-adaptive trees, and in the latter case, it performs much worse compared to ERM. Finally, we remark that our lower bounds when f^* does not satisfy MSP also hold more broadly for RFs, as well as for trees and ensembles grown with other greedy recursive partitioning strategies.[4]

1.3. *Marginal signal bottleneck*. When a regression function f^* does not satisfy MSP, its Fourier decomposition must contain a term $\alpha_{S_i}\chi_{S_i}$ that violates (5). This term creates problems for CART in a similar way as the XOR function. Let us call a covariate X_k *marginally undetectable* in $\mathcal{C}$ if (3) holds. As argued before, when X_k is relevant but marginally undetectable, CART cannot distinguish it from irrelevant covariates. One can show that there exists covariate indices $k_1, k_2 \in S_i$ such that X_{k_1} is marginally undetectable unless $\mathcal{C}$ has already has split on x_{k_2} and X_{k_2} is marginally undetectable unless $\mathcal{C}$ has already has split on x_{k_1}. We call this chicken and egg problem *marginal signal bottleneck*. It results in a highly probable event in which both these covariates remain marginally undetectable at every iteration of CART, leading to high-dimensional inconsistency.

On the other hand, it is easy to show that MSP together with genericity implies Chi et al. [18]'s sufficient impurity decrease (SID) condition. In the setting of uniform binary features, the SID condition asserts a lower bound for what we call the *marginal signal*,

$$\max_{1 \le k \le d} \mathrm{Corr}^2\{f^*(\mathbf{X}), X_k \mid \mathbf{X} \in \mathcal{C}\} \ge \lambda, \tag{7}$$

that holds uniformly over all subcubes $\mathcal{C}$ on which $\mathrm{Var}\{f^*(\mathbf{X}) | \mathbf{X} \in \mathcal{C}\} > 0$. When (7) holds for *some* $\mathcal{C}$, then with enough samples, CART is able to identify and select a relevant covariate for splitting $\mathcal{C}$. When (7) hold for *all* subcubes (i.e., SID), we are able to guarantee that CART is uniformly able to identify and select relevant covariates across every iteration loop of the algorithm, thereby yielding high-dimensional consistency.

[4]It is possible to extend the upper bound to various forms of RFs, but we leave this to future work.

1.4. *Interpreting CART as a stochastic process.* Despite the simplicity of the lower bound argument outlined in the previous section, making it rigorous in finite samples, even in the special case of the XOR function, has proved elusive until now. To illustrate the difficulty, consider a fixed query point $\mathbf{x} \in \{\pm 1\}^d$, and consider $\mathcal{Q}$, the root-to-leaf path taken by $\mathbf{x}$. We have to show that there is a highly probable event on which $\mathcal{Q}$ does not split on x_1 or x_2. The usual approach to such a task is to attempt to prove concentration of the split criteria (2) over all nodes (subcubes) along the path. However, because of the data adaptivity of CART splits, we have little control over which subcubes appear, and we would be forced to derive concentration bounds that are uniform over all subcubes. Unfortunately, this desired result is too strong and does not hold.

To resolve this, we interpret $\mathcal{Q}$ as a stochastic process. The measurement at "time point" t comprises the values $\mathrm{Corr}^2\{f^*(\mathbf{X}), X_k \mid \mathbf{X} \in \mathcal{C}_t\}$, $k = 1 \ldots, d$, where $\mathcal{C}_t$ is the node in $\mathcal{Q}$ at depth t. We next couple this stochastic process to one that makes totally random splits, which allows us to use symmetry to calculate the probability that it does not split on x_1 or x_2. On the event that the two processes are coupled, this also holds for the CART path $\mathcal{Q}$.

Although we have outlined a lower bound for the XOR function, it can be readily adapted to any function not satisfying MSP. Furthermore, there is nothing particularly special about the CART criterion. The proof relies on the fact that splits are determined by considering only the marginal distributions $X_k | \mathcal{C}, Y$ for $k = 1, \ldots, d$ (as opposed to the joint distribution $\mathbf{X}| \mathcal{C}, Y$), and hence extends to a larger class of greedy recursive partitioning algorithms, which we formally define in Section 2. As far as we are aware, the proof technique is completely novel.

1.5. *Robust lower bounds under 'soft" bottlenecks.* Thus far, we have seen that when f^* does not satisfy MSP, it experiences marginal signal bottleneck, which leads to high-dimensional inconsistency. In some sense, however, being non-MSP is not a robust property because a small perturbation of the zero coefficients of such a function results in one that does satisfy MSP. As such, our argument regarding the weaknesses of greedy recursive partitioning strategies would be stronger if we could show that their performance degrades smoothly as the regression function f^* approaches the class of non-MSP functions. Fortunately, this is indeed the case, as reflected by the following informal theorem.

THEOREM 1.2 (Informal). *When $\mathcal{X} = \{\pm 1\}^d$, $\nu = \mathrm{Unif}(\{\pm 1\}^d)$, f^* depends only on s covariates, then the expected L^2 risk of the CART estimator $\hat{f}_{\mathrm{CART}}$ satisfies*

$$E_{\mathcal{D}_n}\{R(\hat{f}(-;\mathcal{D}_n), f^*)\} = \Omega(1), \quad \text{whenever } n = \min\Big\{O\Big(1/\min_{f \notin \mathsf{MSP}} R(f, f^*)\Big), \exp\big(O(d)\big)\Big\}.$$

The value, $\min_{f \notin \mathsf{MSP}} R(f, f^*)$, can be thought of as measuring the width of a "soft" bottleneck. For example, if we set $f^*(\mathbf{x}) = \alpha x_1 + x_2 + x_1 x_2 x_3$, then we have $\min_{f \notin \mathsf{MSP}} R(f, f^*) = \alpha^2$. We thus see that the sample complexity lower bound scales inversely with the bottleneck width, thereby extending the performance gap between greedy and ERM trees to a larger range of settings. Finally, we remark that MSP can be re-interpreted in graph theoretic terms as form of connectedness. Under this interpretation, $\min_{f \notin \mathsf{MSP}} R(f, f^*)$ is equal to the weight of the minimum vertex cut. We elaborate upon this perspective in Section 5.

1.6. *Comparisons with neural networks trained by SGD.* We now compare and connect our results with the literature on training neural networks. Readers who are only interested in decision trees and ensembles can safely skip this subsection.

In computational learning theory, the problem of "learning" (i.e., achieving small estimation error for) sparse Boolean monomials, also called *parities*, from i.i.d. noiseless observations (i.e., $\varepsilon_i = 0$ in (1)) is known to be statistically easy but computationally hard [10], and hence has served as a useful benchmark for learning algorithms and computational frameworks. Recently, this has become a benchmark for studying neural networks (NNs) trained using SGD, which when combined with our results, allows us to perform, to our knowledge, the first theoretical head-to-head comparison between this class of algorithms and greedy regression trees and ensembles.

Indeed, the MSP was introduced by [1] to generalize Boolean monomials. They showed that in the mean-field regime (very wide two-layer NNs with very small step size; see also [38]), Cd iterations of online SGD (one sample per iteration) are sufficient (for some C) to achieve small estimation error whenever f^* satisfies MSP and is generic but is insufficient (for any C) when f^* does not satisfy MSP. Intuitively, functions that do not satisfy MSP produce optimization landscapes with saddle points, which mean-field online SGD struggles to escape from. Comparing with Theorem 1.1, this creates an interesting analogy between mean-field online SGD and greedy regression trees and ensembles.

On the other hand, later works inadvertently emphasized the *differences* between greedy regression trees and ensembles and NNs trained outside the mean-field regime and beyond the $O(d)$ iteration horizon. In the classification setting, [24] showed that online minibatch SGD on a two-layer NN can learn the XOR function with $\tilde{\Theta}(d)$ samples and iterations. Kou et al. [34] showed that for any k, online *sign*-SGD with a batch size of $\tilde{O}(d^{k-1})$ can learn a k-parity (i.e., degree k monomial) within $O(\log d)$ iterations, for a total sample complexity of $\tilde{O}(d^{k-1})$. Since our lower bound in Theorem 1.1 holds even with noiseless observations, this establishes a rigorous performance gap between the two algorithm classes for some non-MSP functions.

Of further interest is [2]'s conjecture characterizing the sample complexity required to learn any given Boolean function in terms of its "leap complexity." If this conjecture is true, NNs trained with SGD outperform greedy regression trees and ensembles on all non-MSP functions but perform more poorly in comparison on MSP functions. We discuss these comparisons more elaborately in Section K of the Supplementary Material.

2. Regression trees.

In this section, we formally define regression tree models. We do so in the context of binary covariates, which simplifies both the definitions of these objects and the notation required to describe them. The reader is referred to [26] on how to adapt these definitions to a more general context.

2.1. *Cells, partitions and splits.*

A *cell* $\mathcal{C}$ is a subcube of $\{\pm 1\}^d$ and is defined as

$$\mathcal{C} = \{\mathbf{x} \in \{\pm 1\}^d : x_j = z_j \text{ for } j \in J(\mathcal{C})\} \tag{8}$$

for a subset of covariate indices $J(\mathcal{C}) \subset [d]$ and a sequence of signs $z_j \in \{\pm 1\}$, $j \in J(\mathcal{C})$. Given a choice of covariate index $k \notin J(\mathcal{C})$ and sign $z \in \{\pm 1\}$, we define the *split operator* as producing the output $\mathcal{T}_{k,z}(\mathcal{C}) := \{\mathbf{x} \in \mathcal{C} : x_k = z\}$. $\mathcal{T}_{k,1}(\mathcal{C})$ and $\mathcal{T}_{k,-1}(\mathcal{C})$ are called the *right* and *left* child of cell $\mathcal{C}$. A *partition* $\mathcal{P}$ is a collection of disjoint cells whose union comprises $\{\pm 1\}^d$. The *size* of $\mathcal{P}$ is the number of cells it contains and is denoted $|\mathcal{P}|$. We say that another partition $\mathcal{P}'$ is a *refinement* of $\mathcal{P}$, denoted $\mathcal{P} \subset \mathcal{P}'$, if for any $\mathcal{C}' \in \mathcal{P}'$, there exists $\mathcal{C} \in \mathcal{P}$ such that $\mathcal{C}' \subset \mathcal{C}$. We say that $\mathcal{P}'$ is a *simple refinement* of $\mathcal{P}$, denoted $\mathcal{P} \subset_s \mathcal{P}'$ if $\mathcal{P} \backslash \mathcal{P}' = \{\mathcal{C}\}$, $\mathcal{P}' \backslash \mathcal{P} = \{\mathcal{C}'_L, \mathcal{C}'_R\}$ and $\mathcal{C} = \mathcal{C}'_L \cup \mathcal{C}'_R$. In other words, $\mathcal{P}'$ is obtained from $\mathcal{P}$ by applying the split operator to $\mathcal{C}$. We use $I(\mathcal{C})$ to denote the indices of observations whose covariate vectors lie in $\mathcal{C}$, that is, $I(\mathcal{C}) := \{i \in [n] : \mathbf{x}_i \in \mathcal{C}\}$, and let $N(\mathcal{C}) := |I(\mathcal{C})|$. Furthermore, we denote the mean response in $\mathcal{C}$ as $\bar{Y}_{\mathcal{C}} := \frac{1}{N(\mathcal{C})} \sum_{i \in I(\mathcal{C})} Y_i$.

2.2. *Regression tree models.* Note that every piecewise constant function g on $\{\pm 1\}^d$ can be parameterized using a partition $\mathcal{P}$ with a fixed ordering of its constituent cells $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_{|\mathcal{P}|}$, and a set of labels $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_{|\mathcal{P}|})$ representing the values taken by g on each cell. In other words, we have

$$g(\mathbf{x}; \mathcal{P}, \boldsymbol{\mu}) = \sum_{j=1}^{|\mathcal{P}|} \mu_j \mathbb{1}\{\mathbf{x} \in \mathcal{C}_j\}. \tag{9}$$

A binary, axis-aligned *regression tree model* is a special type of piecewise constant function, one whose partition $\mathcal{P}$ arises from recursive applications of split operators $\mathcal{T}_{\cdot,\cdot}$ to $\{\pm 1\}^d$ and the resulting cells. More precisely, there exists a sequence of simple refinements, $\{\{\pm 1\}^d\} = \mathcal{P}_1 \subset_s \mathcal{P}_2 \subset_s \cdots \subset_s \mathcal{P}_{|\mathcal{P}|} = \mathcal{P}$. This recursive structure can be represented as a labeled binary tree $\mathfrak{T}$ in which each node corresponds to a cell obtained in the recursion and parent-child relationships correspond to applications of $\mathcal{T}_{\cdot,\cdot}$. The *root* of $\mathfrak{T}$ is the entire covariate space $\{\pm 1\}^d$. As is common in the literature, we will abuse terminology and sometimes use the term "node" to refer to its corresponding cell. Each internal node is labeled with the covariate used to split it, called its *split covariate* and denoted as $k(\mathcal{C})$. The non-internal nodes $\mathcal{L}_1, \mathcal{L}_2, \ldots, \mathcal{L}_{|\mathcal{P}|}$ are called *leaves*, have empty labels and comprise the final partition $\mathcal{P} = \mathcal{P}(\mathfrak{T})$. For convenience, we also slightly abuse notation, writing $|\mathfrak{T}| = |\mathcal{P}(\mathfrak{T})|$ and $g(-; \mathfrak{T}, \boldsymbol{\mu}) = g(-; \mathcal{P}(\mathfrak{T}), \boldsymbol{\mu})$. We call $\mathfrak{T}$ the *tree structure* of the model and $\boldsymbol{\mu}$ its *leaf labels*. The *depth* of $\mathfrak{T}$ is the maximum length of a *root-to-leaf path*, that is, a sequence of nodes $\mathcal{C}_0 \supset \mathcal{C}_1 \supset \cdots \supset \mathcal{C}_l$, where $\mathcal{C}_j$ is a parent of $\mathcal{C}_{j+1}$ in $\mathfrak{T}$ for $j = 0, 1, \ldots, l-1$, $\mathcal{C}_0$ is the root and $\mathcal{C}_l$ is a leaf node.

2.3. *ERM tree models.* Given the observations $\mathcal{D}_n$, the L^2 empirical risk of a regression function $f \colon \{\pm 1\}^d \to \mathbb{R}$ is defined as

$$\hat{R}(f; \mathcal{D}_n) := \frac{1}{n} \sum_{i=1}^{n} \big(Y_i - f(\mathbf{X}_i)\big)^2. \tag{10}$$

The empirical risk minimizer or *ERM tree model* is the regression tree model minimizing this functional subject to a constraint set Γ on the tree structure and upper bound M on the possible leaf label values:

$$\hat{f}_{\mathrm{ERM}}(-; \mathcal{D}_n) := \operatorname*{arg\,min}_{\mathfrak{T} \in \Gamma,\, \|\boldsymbol{\mu}\|_\infty \le M} \hat{R}\big(g(-; \mathfrak{T}, \boldsymbol{\mu}); \mathcal{D}_n\big). \tag{11}$$

Note that for any fixed tree, the leaf labels that minimize the empirical risk are exactly the mean responses $\overline{Y}_{\mathcal{L}_j}$, so long as these values do not exceed M in absolute value.

We include the constraint $\|\boldsymbol{\mu}\|_\infty \le M$ in (11) in order to obtain a uniform L_∞ envelope for the class of tree functions. This is used in Proposition 6.2, where we apply local Gaussian and Rademacher complexity bounds to derive a high-probability excess-risk bound of order $\sigma^2 + M^2$. Without such an envelope, the relevant complexities can be unbounded, and the resulting nonasymptotic risk bound may fail to be finite. In typical settings where Y is bounded or sub-Gaussian, one can moreover choose M of order $\|f^*\|_\infty + \sigma\sqrt{\log n}$ so that, with high probability, the unconstrained ERM and its truncated version coincide. One can also potentially use the approach by [39] that avoids the aforementioned restrictive assumptions. But carrying out the details is truly beyond the scope of this current work.

2.4. *Regression tree algorithms.* For computational reasons, regression tree models are usually defined as the output of an algorithm rather than as the solution to an ERM problem (11) or any other global optimization program. We call such an algorithm a *regression tree* *algorithm*, treating it as a function $\mathcal{A}$ that takes in the data $\mathcal{D}_n$ together with a random seed Θ and returns a regression tree model $g(-;\mathfrak{T},\boldsymbol{\mu}) = \mathcal{A}(\mathcal{D}_n,\Theta)$. Since g is fully determined by its tree structure and leaf labels, we can write the algorithm as a tuple $\mathcal{A} = (\mathcal{A}_t,\mathcal{A}_l)$, where $\mathfrak{T} = \mathcal{A}_t(\mathcal{D}_n,\Theta)$ and $\boldsymbol{\mu} = \mathcal{A}_l(\mathcal{D}_n,\Theta,\mathcal{A}_t(\mathcal{D}_n,\Theta))$. We call $\mathcal{A}_t$ and $\mathcal{A}_l$ the tree structure component and leaf labels component of the algorithm respectively. In most algorithms used in practice, $\mathcal{A}_l$ sets the label of each leaf to be the mean response within that leaf. Notwithstanding recent work commenting about the possible limitations of this approach [6, 46], we hence assume this to be the case unless stated otherwise, thereby putting the focus squarely on the tree structure component $\mathcal{A}_t$.

2.5. *CART*. Breiman et al.'s [13] *CART* is a regression tree algorithm that grows a tree greedily in a top-down iterative fashion as follows: Starting with the entire space $\{\pm1\}^d$, each new cell $\mathcal{C}$ obtained in the algorithm is first checked for a stopping condition. If the condition is met, $\mathcal{C}$ is not split and is fixed as a leaf in the final tree. Otherwise, its split covariate is selected to be the maximizer of the *impurity decrease*, which is defined as

$$\begin{aligned}\hat{\Delta}(k;\mathcal{C},\mathcal{D}_n) := \mathcal{I}(\mathcal{C};\mathcal{D}_n) &- \frac{N(\mathcal{T}_{k,1}(\mathcal{C}))}{N(\mathcal{C})}\mathcal{I}\big(\mathcal{T}_{k,1}(\mathcal{C});\mathcal{D}_n\big)\\ &- \frac{N(\mathcal{T}_{k,-1}(\mathcal{C}))}{N(\mathcal{C})}\mathcal{I}\big(\mathcal{T}_{k,-1}(\mathcal{C});\mathcal{D}_n\big),\end{aligned} \tag{12}$$

where $\mathcal{I}(\mathcal{C};\mathcal{D}_n) := \frac{1}{N(\mathcal{C})}\sum_{i\in I(\mathcal{C})}(Y_i - \bar{Y}_{\mathcal{C}})^2$ is called the *impurity* of the cell $\mathcal{C}$.

The term "impurity" here refers to Gini impurity, which is related to the probability of misclassification in classification trees. In a regression tree setting, we see that it is exactly equal to the empirical variance of the response within the cell. The second and third terms on the right-hand side in (12) sum to the residual variance in $\mathcal{C}$ after regressing out X_k, which means that the impurity decrease can be written as

$$\hat{\Delta}(k;\mathcal{C},\mathcal{D}_n) = \mathcal{I}(\mathcal{C};\mathcal{D}_n)\cdot\widehat{\operatorname{Corr}}^2\{Y, X_k \mid \mathbf{X}\in\mathcal{C}\} \tag{13}$$

as alluded to earlier.

When all leaves satisfy the stopping condition, the current tree $\mathfrak{T}$ is sometimes pruned to a subtree $\mathfrak{T}' \subset \mathfrak{T}$. $\mathfrak{T}'$ is then returned as the output to the algorithm. Several different stopping conditions and pruning rules are available [13, 42]. Our lower bounds will not depend on these choices so we will not go into further detail on what these are. Finally, for our upper bounds, we will make use of the *minimum impurity decrease* stopping condition, which stops splitting $\mathcal{C}$ when it satisfies

$$\frac{N(\mathcal{C})}{n}\max_{k\in[d]\backslash J(\mathcal{C})}\hat{\Delta}(k;\mathcal{C},\mathcal{D}_n) < \gamma \tag{14}$$

for some prespecified threshold γ. We call an output of the algorithm a *CART model*, and denote it as $\hat{f}_{\mathrm{CART}}(-;\mathcal{D}_n)$.

2.6. *Random forests*. A *random forest* model is an ensemble of randomized CART trees. Specifically, for each CART tree, we fit it to a bootstrapped version of the dataset $\mathcal{D}_n$, denoted $\mathcal{D}_n^*$ and furthermore, at each iteration, a random subset $\mathcal{M}\subset[d]$ of some prespecified size is drawn uniformly at random with the split covariate is selected from among these covariates, that is, we set

$$k(\mathcal{C}) := \operatorname*{arg\,max}_{k\in\mathcal{M}\backslash J(\mathcal{C})}\hat{\Delta}\big(k;\mathcal{C},\mathcal{D}_n^*\big). \tag{15}$$

Slightly abusing notation, we use $\hat{f}_{\mathrm{CART}}(-;\mathcal{D}_n,\theta)$ to denote the CART model fitted with a random seed θ. A random forest model with M trees is then

$$\hat{f}_{\mathrm{RF}}(-;\mathcal{D}_n,\Theta) := \frac{1}{M}\sum_{m=1}^{M}\hat{f}_{\mathrm{CART}}(-;\mathcal{D}_n,\theta_m), \tag{16}$$

where $\Theta = (\theta_1,\theta_2,\dots,\theta_M)$ and $\theta_1,\theta_2,\dots,\theta_M$ are drawn i.i.d.

2.7. *Greedy tree models.* In order to isolate the aspect of CART that leads to its limitations with non-MSP functions, we make a more general definition. We say that $\mathcal{A}$ is a *greedy tree algorithm* if, similar to CART, it grows a tree greedily in a top–down fashion, but uses a possibly different criterion $\mathcal{O}(k;\mathcal{C},\mathcal{D}_n)$ instead of impurity decrease (12) to determine its splits. We constrain $\mathcal{O}(k;\mathcal{C},\mathcal{D}_n)$ to depend on $\mathcal{D}_n$ only via the values of the kth covariate and the response for observations in $\mathcal{C}$, that is,

$$\mathcal{O}(k;\mathcal{C},\mathcal{D}_n) = \mathcal{F}\big(\{(X_{ik},Y_i)\}_{i\in I(\mathcal{C})}\big) \tag{17}$$

for a fixed function $\mathcal{F}$ that is invariant under the map $\{(X_{ik},Y_i)\}_{i\in I(\mathcal{C})} \mapsto \{(-X_{ik},Y_i)\}_{i\in I(\mathcal{C})}$.[5] Since

$$\mathcal{I}(\mathcal{T}_{k,z}(\mathcal{C});\mathcal{D}_n) = \sum_{i\in I(\mathcal{C})}\left(\frac{|X_{ik}+z|}{2}Y_i - \frac{\sum_{i\in I(\mathcal{C})}|X_{ik}+z|Y_i}{\sum_{i\in I(\mathcal{C})}|X_{ik}+z|}\right)^2, \tag{18}$$

for $z=-1,1$, we see that CART is indeed an instance of a greedy tree algorithm. We call an output of $\mathcal{A}$ a *greedy tree model* and denote it as $\hat{f}_{\mathrm{G}}$. Furthermore, if $\hat{f}_{\mathrm{G}}$ is the base learner in an ensemble constructed analogously to a random forest, we call the resulting ensemble a *greedy tree ensemble* and denote it as $\hat{f}_{\mathrm{G\text{-}E}}$.

3. Covariate selection and high-dimensional inconsistency.

3.1. *High-dimensional consistency.* We will compare and contrast the performance of ERM trees and greedy trees in a sparse, high-dimensional setting. To formalize this setting, we make the following assumption for the rest of this paper.

ASSUMPTION 3.1. Assume a binary covariate space $\mathcal{X} = \{\pm 1\}^d$ with ν the uniform measure. Let $f_0^*\colon \{\pm 1\}^s \to \mathbb{R}$ be any function. Let $S^* \subset [d]$ be a subset of size $|S^*| = s$. We consider the nonparametric model (1) with the regression function $f^*\colon \{\pm 1\}^d \to \mathbb{R}$ defined via $f^*(\mathbf{x}) = f_0^*(\mathbf{x}^{S^*})$.

We evaluate the performance of an estimator $\hat{f}$ for f^* using the expected L^2 risk, where the expectation is taken over the randomness with respect to $\mathcal{D}_n$ as well as potential algorithmic randomness Θ:

$$\mathfrak{R}(\hat{f}, f_0^*, d, n) := \mathbb{E}_{\mathcal{D}_n,\Theta}\big\{R\big(\hat{f}(-;\mathcal{D}_n,\Theta), f^*\big)\big\}. \tag{19}$$

We say that $\hat{f}$ has *high-dimensional consistency* with respect to f_0^* if there is an infinite sequence $d_1, d_2, \dots$ such that $\lim_{n\to\infty}\mathfrak{R}(\hat{f}, f_0^*, d_n, n) = 0$ and $\lim_{n\to\infty}\log n/d_n = 0$. In other words, estimators that are not high-dimensional consistent require, in order to estimate f^*, a number of samples that grows exponentially in the ambient dimension. Note that this definition is intentionally weak, so as to emphasize the weakness of greedy trees under non-MSP conditions.

[5]This condition is used in the proof of Lemma C.1.

3.2. *Reduction to covariate selection along query paths.* As argued heuristically in the Introduction, the ability to perform some sort of covariate selection is necessary for high-dimensional consistency. We begin to make this intuition rigorous by introducing the notion of a *query path*. Given a choice of (i) a query point $\mathbf{x} \in \{\pm 1\}^d$, (ii) a regression tree structure algorithm $\mathcal{A}_t$, (iii) a dataset $\mathcal{D}_n$ and (iv) a random seed Θ for $\mathcal{A}_t$, there is a unique leaf $\mathcal{L}$ in the tree structure $\mathcal{A}_t(\mathcal{D}_n, \Theta)$ that contains $\mathbf{x}$. We call the path in $\mathcal{A}_t(\mathcal{D}_n, \Theta)$ from the root to $\mathcal{L}$ the query path of $\mathbf{x}$, and denote it using $\mathcal{Q}(\mathbf{x}; \mathcal{D}_n, \Theta)$. Of special interest to us will be the covariates split on along the query path, that is,

$$J(\mathbf{x}; \mathcal{D}_n, \Theta) := \{k(\mathcal{C}) : \mathcal{C} \in \mathcal{Q}(\mathbf{x}; \mathcal{D}_n, \Theta)\}. \tag{20}$$

Note that to simplify notation, we have elided the dependence of $\mathcal{A}_t$, whose choice will be clear from the context. The role of the query path in expected risk lower bounds is established in the following two lemmas.

LEMMA 3.2. *Let $\hat{f}$ be a regression tree model fit using a regression tree algorithm $\mathcal{A}$. Then under Assumption 3.1, its expected L^2 risk satisfies $\mathfrak{R}(\hat{f}, f_0^*, d, n) \geq (1-\delta)\operatorname{Var}\{f_0^*(\mathbf{X})\}$, where $\delta := \mathbb{P}\{J(\mathbf{X}; \mathcal{D}_n, \Theta) \cap S^* \neq \varnothing\}$.*

PROOF. We start with the trivial lower bound

$$R(\hat{f}(-; \mathcal{D}_n, \Theta), f^*) \geq \mathbb{E}_{\mathbf{X}\sim\nu}\{(\hat{f}(\mathbf{X}; \mathcal{D}_n, \Theta) - f^*(\mathbf{X}))^2 \mathbb{1}\{J(\mathbf{X}; \mathcal{D}_n, \Theta) \cap S^* = \varnothing\}\}. \tag{21}$$

The collection $\{J(\mathbf{x}; \mathcal{D}_n, \Theta) \cap S^* = \varnothing\}_{\mathbf{x}\in\{\pm1\}^d}$ can be thought of as a collection of random events indexed by $\mathbf{x} \in \{\pm 1\}^d$. Whenever one of these events holds for some fixed value $\mathbf{x}$, $\hat{f}(\mathbf{x}; \mathcal{D}_n, \Theta)$ is constant with respect to changes in $\mathbf{x}_{S^*}$, while by definition $f^*(\mathbf{x})$ is constant with respect to changes in $\mathbf{x}_{[d]\setminus S^*}$. Utilizing these facts together with the uniform distribution of $\mathbf{X}$ allows us to decompose the conditional expectation of the squared loss as

$$\begin{aligned} &\mathbb{E}_{\mathbf{X}}\{(\hat{f}(\mathbf{X}; \mathcal{D}_n, \Theta) - f^*(\mathbf{X}))^2 \mid \mathbf{X}_{[d]\setminus S^*} = \mathbf{x}_{[d]\setminus S^*}\} \\ &\quad = \operatorname{Var}\{f_0^*(\mathbf{X})\} + (\hat{f}(\mathbf{x}; \mathcal{D}_n, \Theta) - \mathbb{E}\{f_0^*(\mathbf{X})\})^2 \\ &\quad \geq \operatorname{Var}\{f_0^*(\mathbf{X})\}. \end{aligned} \tag{22}$$

Making the further observation that the function $\mathbb{1}\{J(\mathbf{x}; \mathcal{D}_n, \Theta) \cap S^* = \varnothing\}$ is constant with respect to $\mathbf{x}_{S^*}$, we can use properties of conditional expectations to rewrite the right-hand side of (21) as

$$\mathbb{E}_{\mathbf{X}\sim\nu}\{\mathbb{E}_{\mathbf{X}}\{(\hat{f}(\mathbf{X}; \mathcal{D}_n, \Theta) - f^*(\mathbf{X}))^2 \mid \mathbf{X}_{[d]\setminus S^*}\}\mathbb{1}\{J(\mathbf{X}; \mathcal{D}_n, \Theta) \cap S^* = \varnothing\}\}. \tag{23}$$

Plugging in the bound (22), taking a further expectation with respect to $\mathcal{D}_n$ and Θ, and simplifying the resulting expression completes the proof. □

LEMMA 3.3. *Let $\hat{f}$ be a regression tree ensemble model in which each tree is fit using a regression tree algorithm $\mathcal{A}$. In addition to Assumption 3.1, assume that the response variable is bounded almost surely, that is, $|Y| \leq M$. Then its expected L^2 risk satisfies $\mathfrak{R}(\hat{f}, f_0^*, d, n) \geq (1-\kappa\delta)\operatorname{Var}\{f_0^*(\mathbf{X})\}$, where $\delta := \max_{\mathbf{x}} \mathbb{P}\{J(\mathbf{x}; \mathcal{D}_n, \Theta) \cap S^* \neq \varnothing\}$ and $\kappa := 2M/\operatorname{Var}\{f_0^*(\mathbf{X})\}^{1/2}$.*

PROOF. We first write

$$\hat{f}(\mathbf{x}; \mathcal{D}_n, \Theta) = \frac{1}{B}\sum_{b=1}^{B} \tilde{f}(\mathbf{x}; \mathcal{D}_n, \theta_b), \tag{24}$$

where $\tilde{f}$ is the base learner of the ensemble. Taking expectations with respect to $\mathcal{D}_n$ and Θ gives the chain of equalities:

$$
\begin{aligned}
\mathbb{E}_{\mathcal{D}_n,\Theta}\left\{\frac{1}{B}\sum_{b=1}^{B}\tilde{f}(\mathbf{x};\mathcal{D}_n,\theta_b)\right\} &= \mathbb{E}_{\mathcal{D}_n,\theta_1}\{\tilde{f}(\mathbf{x};\mathcal{D}_n,\theta_1)\} \\
&= \underbrace{\mathbb{E}_{\mathcal{D}_n,\theta_1}\{\tilde{f}(\mathbf{x};\mathcal{D}_n,\theta_1)\mathbb{1}\{J(\mathbf{x};\mathcal{D}_n,\theta_1)\cap S^*=\varnothing\}\}}_{g(\mathbf{x})} \\
&\quad + \underbrace{\mathbb{E}_{\mathcal{D}_n,\theta_1}\{\tilde{f}(\mathbf{x};\mathcal{D}_n,\theta_1)\mathbb{1}\{J(\mathbf{x};\mathcal{D}_n,\theta_1)\cap S^*\neq\varnothing\}\}}_{h(\mathbf{x})}.
\end{aligned} \tag{25}
$$

For ease of notation, we label the two quantities on the right-hand side as $g(\mathbf{x})$ and $h(\mathbf{x})$ respectively. Note also that $|h(\mathbf{x})| \leq M\delta$. Using Jensen's inequality followed by the tower property of conditional expectations, we may then compute

$$
\begin{aligned}
&\mathbb{E}_{\mathbf{X},\Theta,\mathcal{D}_n}\{(\hat{f}(\mathbf{X};\mathcal{D}_n,\Theta)-f^*(\mathbf{X}))^2\} \\
&\quad \geq \mathbb{E}_{\mathbf{X}}\{(g(\mathbf{X})+h(\mathbf{X})-f^*(\mathbf{X}))^2\} \\
&\quad = \mathbb{E}_{\mathbf{X}}\{\mathbb{E}_{\mathbf{X}}\{(g(\mathbf{X})+h(\mathbf{X})-f^*(\mathbf{X}))^2 \mid \mathbf{X}_{[d]\setminus S^*}\}\}.
\end{aligned} \tag{26}
$$

We next notice that $g(\mathbf{x})$ is constant with respect to changes in $\mathbf{x}_{S^*}$. This means that, similar to (22), the conditional expectation within (26) can be lower bounded by a variance term, which we expand as follows:

$$
\begin{aligned}
&\mathbb{E}_{\mathbf{X}}\{(g(\mathbf{X})+h(\mathbf{X})-f^*(\mathbf{X}))^2 \mid \mathbf{X}_{[d]\setminus S^*}=\mathbf{x}_{[d]\setminus S^*}\} \\
&\quad \geq \operatorname{Var}_{\mathbf{X}}\{h(\mathbf{X})-f^*(\mathbf{X}) \mid \mathbf{X}_{[d]\setminus S^*}=\mathbf{x}_{[d]\setminus S^*}\} \\
&\quad \geq \operatorname{Var}\{f_0^*(\mathbf{X})\} - 2\operatorname{Var}\{f_0^*(\mathbf{X})\}^{1/2}\operatorname{Var}_{\mathbf{X}}\{h(\mathbf{X}) \mid \mathbf{X}_{[d]\setminus S^*}=\mathbf{x}_{[d]\setminus S^*}\}^{1/2}.
\end{aligned} \tag{27}
$$

We now plug this back into (26) to get

$$
\mathfrak{R}(\hat{f}, f_0^*, d, n) \geq (1 - 2M\delta/\operatorname{Var}\{f_0^*(\mathbf{X})\}^{1/2})\operatorname{Var}\{f_0^*(\mathbf{X})\}. \tag{28}
$$

□

4. Greedy trees and the XOR function. Before presenting our main results in full generality, we first state and discuss how to obtain an expected risk lower bound for the XOR function $f_0^*(x_1, x_2) = x_1 x_2$ in the sparse high-dimensional setting (Assumption 3.1). As mentioned in the Introduction, the XOR function does not contain any marginal signal and hence presents perhaps the simplest setting in which CART has been observed to perform much worse than ERM trees. The example has indeed been discussed before in the literature, but no formal proof regarding this phenomenon was previously available.

PROPOSITION 4.1. *Under Assumption 3.1, let f_0^* be the XOR function. For any $0 < \delta < 1$, suppose $n \leq 2^{\delta(d-1)/2-2}$. Then the expected L^2 risk of any greedy tree model satisfies*

$$
\mathfrak{R}(\hat{f}_{\mathrm{G}}, f_0^*, d, n) \geq (1-\delta)\operatorname{Var}\{f_0^*(\mathbf{X})\}. \tag{29}
$$

If furthermore, $|Y| \leq M$ almost surely, then the L^2 risk for any greedy tree ensemble model satisfies

$$
\mathfrak{R}(\hat{f}_{\text{G-E}}, f_0^*, d, n) \geq (1-\kappa\delta)\operatorname{Var}\{f_0^*(\mathbf{X})\}, \tag{30}
$$

where $\kappa := 2M/\operatorname{Var}\{f_0^*(\mathbf{X})\}^{1/2}$. In particular, $\hat{f}_{\mathrm{G}}$ and $\hat{f}_{\text{G-E}}$ are high-dimensional inconsistent for f_0^*.

To prove this statement, given Lemma 3.2 and Lemma 3.3, we need only provide an appropriate upper bound for the covariate selection probability along a query path.

LEMMA 4.2. *Under Assumption 3.1, let f_0^* be the XOR function and suppose $\mathcal{A}_t$ is greedy. Then for each fixed query point $\mathbf{x} \in \{\pm 1\}^d$ and random seed Θ, the probability of covariate selection is upper bounded as $\mathbb{P}_{\mathcal{D}_n}\{J(\mathbf{x}; \mathcal{D}_n, \Theta) \cap S^* \neq \varnothing\} \leq 2(\log_2 n + 2)/(d-1)$.*

PROOF. Without loss of generality, assume that $S^* = \{1, 2\}$. The proof idea is to view the query path $\mathcal{Q}(\mathbf{x}; \mathcal{D}_n, \Theta)$ as a stochastic process indexed by node depth. In other words, writing the path as $\mathcal{Q}(\mathbf{x}; \mathcal{D}_n, \Theta)$ and the sequence $\mathcal{C}_0 \supset \mathcal{C}_1 \supset \cdots \supset \mathcal{C}_l$, the information at time t, for $t = 0, 1, \ldots$ comprises the node $\mathcal{C}_t$ as well as its auxiliary information: the split criterion values $\mathcal{O}(k; \mathcal{C}_t, \mathcal{D}_n)$ for $k \notin J(\mathcal{C}_t)$ and split covariate $k_t(\mathbf{x}; \mathcal{D}_n, \Theta) := k(\mathcal{C}_t)$. We will couple this process to two other query path processes, both of which can be shown to select covariates completely at random. On the high probability event that the coupling holds, we are thus able to bound the covariate selection probability of the original query path.

Step 1: Defining the path coupling. Define the modified dataset $\mathcal{D}_n^{(1)} := \{(\mathbf{X}_{i,-1}, Y_i)\}_{i=1}^n$, that is, so that the first covariate is dropped from each observation. Define $\mathcal{D}_n^{(2)}$ analogously, but by dropping X_2 instead of X_1. We will analyze the query paths, $\mathcal{Q}(\mathbf{x}; \mathcal{D}_n^{(1)}, \Theta)$ and $\mathcal{Q}(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta)$, obtained from these modified datasets and compare them with the original.

Step 2: Coupled paths select covariates randomly. For $a = 1, 2$, observe that for any fixed value of $\mathbf{x}_{-a}$, we have

$$\mathbb{P}\{X_1 X_2 = 1 \mid \mathbf{X}_{-a} = \mathbf{x}_{-a}\} = \frac{1}{2}. \tag{31}$$

This implies that the response Y is independent of the covariates in the modified dataset $\mathcal{D}_n^{(a)}$. Combining this with the fact that the covariates are drawn uniformly from $\{\pm 1\}^{d-1}$ as well as the symmetry of the splitting criterion, we see that the distribution of $\mathcal{D}_n^{(a)}$ is invariant to permutation of the covariate indices. As such, $J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta)$ is the prefix of a uniformly random permutation of $[d]\backslash\{a\}$.

Step 3: Bounding covariate selection in coupled paths. For any $b \in [d]\backslash\{a\}$, we use the previous step to compute

$$P\{b \in J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta) \mid |J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta)|\} \leq \frac{|J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta)|}{d-1}. \tag{32}$$

Taking a further expectation and using Lemma C.2 gives

$$\begin{aligned} \mathbb{P}\{b \in J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta)\} &= \mathbb{E}\{\mathbb{P}\{b \in J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta) \mid |J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta)|\}\} \\ &\leq \frac{\log_2 n + 2}{d-1}. \end{aligned} \tag{33}$$

Step 4: Decoupling implies covariate selection. We claim the following inclusion of events:

$$\bigcup_{a=1}^{2} \{J(\mathbf{x}; \mathcal{D}_n, \Theta) \neq J(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta)\} \subset \{1 \in J(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta)\} \cup \{2 \in J(\mathbf{x}; \mathcal{D}_n^{(1)}, \Theta)\}. \tag{34}$$

To prove this, assuming the left-hand side, let t be the smallest index for which $k_t(\mathbf{x}; \mathcal{D}_n^{(a)}, \Theta) \neq k_t(\mathbf{x}; \mathcal{D}_n, \Theta)$ for some a, which we assume without loss of generality is $a = 1$. This implies in particular that the depth t node along all three query paths are equal. Label this node as $\mathcal{C}_t$. We now examine the possible values for $k_t(\mathbf{x}; \mathcal{D}_n, \Theta)$. For $j \neq 1$, we have

$$\mathcal{O}(j, \mathcal{C}_t, \mathcal{D}_n) = \mathcal{F}(\{(X_{ij}, Y_i)\}_{i \in I(\mathcal{C})}) = \mathcal{O}(j, \mathcal{C}_t, \mathcal{D}_n^{(1)}). \tag{35}$$

Recall that $k_t(\mathbf{x}; \mathcal{D}_n, \Theta)$ is defined to be the maximum of the left-hand side over $j \notin J(\mathcal{C}_t)$. Meanwhile, $k_t(\mathbf{x}; \mathcal{D}_n^{(1)}, \Theta)$ is defined to be the maximum of the right-hand side over $j \notin J(\mathcal{C}_t) \cup \{1\}$. As such, we see that the only way in which these can be different values is if $k_t(\mathbf{x}; \mathcal{D}_n, \Theta) = 1$. This would imply, however, that $k_t(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta) = 1$ as we wanted.

Step 5: Completing the proof. Suppose $J(\mathbf{x}; \mathcal{D}_n, \Theta) \cap S^* \neq \varnothing$. If coupling does not hold (i.e., the left-hand side of (34) is true), then we see that either $1 \in J(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta)$ or $2 \in J(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta)$. On the other hand if coupling does hold, then we again have either $1 \in J(\mathbf{x}; \mathcal{D}_n, \Theta) = J(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta)$ or $2 \in J(\mathbf{x}; \mathcal{D}_n, \Theta) = J(\mathbf{x}; \mathcal{D}_n^{(1)}, \Theta)$. In conclusion, we have

$$\begin{aligned} \mathbb{P}_{\mathcal{D}_n}\{J(\mathbf{x}; \mathcal{D}_n, \Theta) \cap S^* \neq \varnothing\} &\leq \mathbb{P}\{1 \in J(\mathbf{x}; \mathcal{D}_n^{(2)}, \Theta)\} + \mathbb{P}\{2 \in J(\mathbf{x}; \mathcal{D}_n^{(1)}, \Theta)\} \\ &\leq \frac{2(\log_2 n + 2)}{d-1}. \end{aligned} \tag{36}$$

□

REMARK 4.3. The technique of controlling a stochastic process by coupling it to another one which allows for more convenient calculations was also used by [49] to analyze the convergence of online SGD for the problem of phase retrieval, and by [3, 5] to analyze gradient descent on leap functions.

5. Greedy trees and non-MSP functions. The XOR function is an example of a function not satisfying [1]'s merged-staircase property (MSP). As promised, the lower bound for the XOR function can be extend to a general lower bound for this class of functions. To motivate why this is the case as well as the type of bounds we may expect, we first further investigate what it means for f^*, written as in (4), to not satisfy MSP.

Construct a directed hypergraph whose vertex set is $\mathcal{S} := \{S_1, S_2, \ldots, S_r\} \cup \varnothing$, in other words, the subsets of $[d]$ corresponding to nonzero Fourier coefficients of f^*, together with the empty set (if not already present). Recall that an edge in a hypergraph is a relationship between a pair of subsets of vertices. In $\mathcal{G}$, we put an edge $\{S_{j_1}, S_{j_2}, \ldots, S_{j_{k-1}}\} \to S_{j_k}$ if

$$\left| S_{j_k} \setminus \bigcup_{i=1}^{k-1} S_{j_i} \right| \leq 1. \tag{37}$$

Finally, put a weight function $w \colon \mathcal{S} \to \mathbb{R}$ defined by $w(S) := \alpha_S^2$. We refer to the weighted graph $\mathcal{G}_{f^*}$ as the *Fourier graph* of f^*.

It is clear that MSP is equivalent to $\mathcal{G}_{f^*}$ being a connected graph. Now let $\mathcal{S}_{\mathsf{MSP}}$ denote the connected component of the empty set $\varnothing$ in $\mathcal{G}_{f^*}$ and set $S^*_{\mathsf{MSP}} := \bigcup_{S \in \mathcal{S}_{\mathsf{MSP}}} S$. By adapting Lemma 4.2, we can show that any covariate X_k with index not in S^*_{MSP} is marginally undetectable, and thus rarely selected, on any query path of a greedy tree. We then modify Lemma 3.2 and Lemma 3.3 to translate lower bounds on the covariate selection probability into more nuanced expected risk lower bounds that can be stated in terms of the MSP residual of f^*, defined as $r_{\mathsf{MSP}} := \sum_{S \in \mathcal{S} \setminus \mathcal{S}_{\mathsf{MSP}}} \alpha_S \chi_S$ (see Lemma D.2 and Lemma D.3 in the Supplementary Material).

Naively following this recipe will produce a result with a sample dependence of $2^{\delta d/s + O(1)}$. In other words, we have a weaker lower bound on the necessary sample size as the support size of f^* increases. This dependence does not always make sense: For instance, the bound obtained for the function $f^* := \chi_{[d]}$ is trivial, even though this is in some sense the most complicated function possible.

To overcome this limitation, we make a new definition. We say a subset $T \subset [d] \backslash S^*_{\mathsf{MSP}}$ is a *traversal* of $\mathcal{S} \backslash \mathcal{S}_{\mathsf{MSP}}$ if for any $S \in \mathcal{S} \backslash \mathcal{S}_{\mathsf{MSP}}$, we have $|T \cap S| \geq 2$. It is easy to see that a traversal always exists.

In other words, a traversal $T \subset [d]$ is a small set of coordinates that intersects each non-MSP interaction in at least two indices. If, along the root-to-leaf path followed by a query point $\mathbf{x}$, a greedy tree never splits on any coordinate in T, then all of these non-MSP interactions remain effectively invisible to the algorithm, and the corresponding residual $r_{\mathsf{MSP}}(\mathbf{x})$ contributes a nonvanishing portion of the risk. Therefore, expected risk lower bounds can be established so long as we can derive lower bounds on the probability that the traversal is selected along a query path (see Lemma D.1 in the Supplementary Material). Putting the pieces together yields the first main result of our paper.

THEOREM 5.1. *Under Assumption 3.1, let $f_0^* = \sum_{S\in\mathcal{S}} \alpha_S \chi_S$ be any non-MSP function, and let $r_{\mathsf{MSP}} := \sum_{S\in\mathcal{S}\backslash\mathcal{S}_{\mathsf{MSP}}} \alpha_S \chi_S$ denote its MSP residual. Let s_{T} be the minimum size of a traversal for $\mathcal{S}\backslash\mathcal{S}_{\mathsf{MSP}}$ and let $s_{\mathsf{MSP}} := |S^*_{\mathsf{MSP}}|$. For any $0 < \delta < 1$, suppose $\log_2 n \le \delta(d - s_{\mathsf{MSP}} - s_{\mathrm{T}} + 1)/s_{\mathrm{T}} - 2$. Then the expected L^2 risk of any greedy tree model satisfies*

$$\mathfrak{R}(\hat{f}_{\mathrm{G}}, f_0^*, d, n) \ge (1-\delta)\operatorname{Var}\{r_{\mathsf{MSP}}(\mathbf{X})\}. \tag{38}$$

If furthermore, $|Y| \le M$ almost surely, then the L^2 risk for any greedy tree ensemble model satisfies

$$\mathfrak{R}(\hat{f}_{\text{G-E}}, f_0^*, d, n) \ge (1-\kappa\delta)\operatorname{Var}\{r_{\mathsf{MSP}}(\mathbf{X})\}, \tag{39}$$

where $\kappa := 2M/\operatorname{Var}\{f_0^*(\mathbf{X})\}^{1/2}$. In particular, $\hat{f}_{\mathrm{G}}$ and $\hat{f}_{\text{G-E}}$ are high-dimensional inconsistent for f_0^*.

PROOF. See Section D of the Supplementary Material. □

REMARK 5.2. It is possible to obtain similar lower bounds that hold with constant probability rather than in expectation. These results are available in Section J of the Supplementary Material.

REMARK 5.3. By orthogonality properties, we have

$$\operatorname{Var}\{r_{\mathsf{MSP}}(\mathbf{x})\} = \sum_{S\in\mathcal{S}\backslash\mathcal{S}_{\mathsf{MSP}}} \alpha_S^2 = w(\mathcal{S}\backslash\mathcal{S}_{\mathsf{MSP}}). \tag{40}$$

In other words, the lower bound value is essentially equal to the total weight of vertices in $\mathcal{G}_{f^*}$ not connected to the empty set. This quantitative connection between lower bounds and Fourier graph properties will be further enriched in Section 8.

As a consequence of Theorem 5.1, we may summarize its implication for the class of greedy tree estimators introduced in Section 2 as follows.

COROLLARY 5.4 (Informal). *Assume $\mathcal{X} = \{\pm 1\}^d$ and $\nu = \operatorname{Unif}(\{\pm 1\}^d)$. Let f^* depend on at most s covariates and suppose that f^* does not satisfy MSP. Then, for any greedy tree estimator $\hat{f}_{\mathrm{G}}$ in the class described in Section 2,*

$$\mathbb{E}_{\mathcal{D}_n}\{R(\hat{f}_{\mathrm{G}}(-;\mathcal{D}_n), f^*)\} = \Omega(1) \quad \textit{whenever } n = \exp(O(d)).$$

In particular, under the high-dimensional scaling regime $\log n = O(d)$, greedy tree procedures are not high-dimensional consistent at non-MSP targets, even when f^* is s-sparse. Together with Proposition 6.2 to come, this yields the statistical–computational trade-off stated in Theorem 1.1: ERM trees achieve vanishing risk in the same regime under the same sparsity assumptions, whereas greedy trees incur a nonvanishing error.

6. ERM trees and minimax rates. To help to contextualize our lower bounds for greedy tree models, we first state minimax lower bounds for the estimation problem defined by Assumption 3.1. It is well known that for sparse regression problems over continuous covariates, consistency is possible under a scaling regime where the sample size n is at most polylogarithmic in the dimension d. This holds both when the regression function is linear [54] and when it is nonlinear [55]. Over binary covariates, standard information theory techniques can be applied to derive a similar minimax lower bound.

PROPOSITION 6.1. *Under Assumption 3.1, further suppose that $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \sim_{i.i.d.} \mathcal{N}(0, \sigma^2)$ and $2^s \log((d-s)/4) \geq 16 \log 2$. Let $\mathfrak{F}_s$ denote all real-valued functions on $\{\pm 1\}^s$. Then we have the minimax lower bound*

$$\inf_{\hat{f}} \max_{f_0^* \in \mathfrak{F}_s} \mathfrak{R}(\hat{f}, f_0^*, d, n) = \Omega\left(\frac{2^s \sigma^2 \log d}{n}\right). \tag{41}$$

PROOF. See Section E of the Supplementary Material. □

Up to logarithmic factors, this minimax rate (in n, d, and s) turns out to be achievable by ERM trees, as shown in the following proposition. In particular, we see that ERM trees are high-dimensional consistent for any f_0^*.

PROPOSITION 6.2. *Under Assumption 3.1, further suppose that $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \sim_{i.i.d.} \mathcal{N}(0, \sigma^2)$ and $\|f_0^*\|_\infty \leq M$. Consider the ERM tree model (11) with the constraint set Γ comprising all tree structures of depth at most s. For any f_0^*, its expected L^2 risk satisfies the upper bound*

$$\mathfrak{R}(\hat{f}_{\mathrm{ERM}}, f_0^*, d, n) = O\left(\frac{(\sigma^2 + M^2)(2^s \log d + \log n)}{n}\right). \tag{42}$$

In particular, $\hat{f}_{\mathrm{ERM}}$ has high-dimensional consistency with respect to f_0^*.

PROOF. See Section F of the Supplementary Material. □

REMARK 6.3. The Gaussian assumption on the noise can be relaxed to a sub-Gaussian assumption. This extension however adds technical complications to the proof and is hence omitted. In comparison, note that our lower bounds for greedy trees and ensembles have not required any additional assumptions on the noise distribution.

7. CART and MSP functions.

7.1. *Characterizing high-dimensional consistency for CART.* In this section, we will show that MSP is a nearly sufficient condition on a regression function f^* in order for CART to have high-dimensional consistency. Together with results from Section 5, this would complete the characterization we desire.

When f^* satisfies MSP, its associated hypergraph is connected. It is easy to verify that this implies that for any cell $\mathcal{C}$ on which f^* is nonconstant, there exists a term $\alpha_S \chi_S$ in its Fourier expansion such that $S \backslash J(\mathcal{C}) = \{j\}$ for some j. Unfortunately, this is not enough to guarantee that X_j is marginally detectable, due to the possibility of cancellations. To see how these may arise, consider the function

$$f^*(\mathbf{x}) = x_1 + x_2 + x_1 x_2 + x_2 x_3. \tag{43}$$

Then on the cell $\mathcal{C} := \{\mathbf{x} \in \{\pm 1\}^d : x_1 = -1\}$, the restriction of f^* to $\mathcal{C}$ satisfies

$$f^*|_{\mathcal{C}}(\mathbf{x}) = -1 + x_2 + (-1)x_2 + x_2x_3 = -1 + x_2x_3. \tag{44}$$

In other words, $f^*|_{\mathcal{C}}$ is non-MSP, and by the results of the previous section, CART as well as other greedy tree methods cannot make further progress.

Fortunately, in some sense, exact cancellations happen rarely. To be more precise, first note that MSP is a condition on the Fourier support $\mathcal{S}$ of f^* rather than on f^* itself. Starting with such a collection, we see that the existence of a cancellation such as that in (44) is a linear constraint on the coefficients $\{\alpha_S\}_{S\in\mathcal{S}}$. If we draw these values from any distribution on $\mathbb{R}^{|\mathcal{S}|}$ that is absolutely continuous with respect to Lebesgue measure, the constraint hence forms a measure zero set, as does the union over all possible constraints. Thus, our results may equivalently be read deterministically for any fixed set of coefficients outside this exceptional set.

We say that a function f^* satisfies the *stable merged-staircase property* (SMSP) if, for any $\mathcal{C} \subset \{\pm 1\}^d$, the restriction $f^*|_{\mathcal{C}}$ is MSP. We have thus shown that under a random perturbation of its coefficients, any MSP function is SMSP almost surely (see Proposition G.1 in Supplementary Material).

Next, one can prove that f^* is SMSP if and only if it satisfies the SID condition (7) for some value of $\lambda > 0$ (see Proposition G.2 in Supplementary Material). Previous work, specifically [18], who first introduced the SID condition, as well as [37], provide upper bounds for the estimation error of CART given the SID condition. These results and proofs are stated for continuous covariates, but can easily be adapted to a discrete covariate setting. We revisit this last statement in the next subsection. Assuming this for now, our discussion thus far yields the following characterization theorem.

THEOREM 7.1. *Let $\mathcal{S}$ be a collection of subsets of S^*. Draw coefficients $\{\alpha_S\}_{S\in\mathcal{S}}$ from any distribution on $\mathbb{R}^{|\mathcal{S}|}$ that is absolutely continuous with respect to Lebesgue measure*, *and set $f_0^* = \sum_{S\in\mathcal{S}} \alpha_S \chi_S$. Then almost surely the following holds under Assumption 3.1*:

- *If $\mathcal{S}$ satisfies* MSP, *then the CART estimator $\hat{f}_{\mathrm{CART}}$ is high-dimensional consistent with respect to f_0^*.*
- *If $\mathcal{S}$ does not satisfy* MSP, *then the CART estimator $\hat{f}_{\mathrm{CART}}$ is high-dimensional inconsistent with respect to f_0^*.*

REMARK 7.2. While [45] obtained a family of upper bounds for CART in a binary covariate setting, their assumed condition (see Assumption 4.1 their paper) turns out to be stronger than the SID condition. Their assumption comprises a type of submodularity for the population impurity decrease values for a *fixed* covariate, more precisely that $\Delta(k; \mathcal{C}) \geq \lambda\Delta(k; \mathcal{C}')$, whenever $\mathcal{C}' \subset \mathcal{C}$ and for all $k \notin J(\mathcal{C}')$. Here,

$$\Delta(k; \mathcal{C}) := \mathrm{Corr}^2\{Y, X_k \mid \mathbf{X} \in \mathcal{C}\} \mathrm{Var}\{f^*(\mathbf{X}) \mid \mathbf{X} \in \mathcal{C}\}. \tag{45}$$

For an example of an SID function that does not satisfy this assumption, consider $f^*(\mathbf{x}) = x_1 + x_1x_2$, in which $\Delta(2, \{\pm 1\}^d) = 0$ and $\Delta(2, \{\mathbf{x} \in \{\pm 1\}^d : x_1 = 1\}) = 1$.

REMARK 7.3. The failure of the directed hypergraph structure $\mathcal{G}_{f^*}$ to fully reflect the statistical properties of the regression function f^* under some circumstances is reminiscent of the potential for DAGs representing a Markov factorization of a distribution to fail to be faithful to its conditional independence structure [51].

REMARK 7.4. It is natural to ask whether our lower bounds can be extended to handle all f^* that do not satisfy SMSP (or equivalently the SID condition). This turns out to be impossible—there exist functions that do not satisfy SMSP, but for which high-dimensional consistency holds. For instance, consider $f^*(\mathbf{x}) = x_1 + 2x_2 + 2x_1x_2 + x_2x_3$. Since $f^*|_{\{x_1=-1\}} = x_2x_3$, this does not satisfy SMSP, but on the other hand, CART would always first split on x_2 given enough samples, in which case this cancellation never occurs. Indeed, by reformulating CART as a sequential greedy approximation algorithm for convex optimization in a Hilbert space, [33] showed high-dimensional consistency for additive models and arbitrary covariates without the SID condition.

7.2. *Sharp upper bounds for CART.* The SID assumption does not always lead to sharp bounds. Consider for instance the sth order AND function $f^*(\mathbf{x}) = \mathbb{1}\{x_j = 1 : j = 1, 2, \dots, s\}$. We may compute $\mathrm{Corr}^2\{f^*(\mathbf{X}), X_j\} = (2^s - 1)^{-1}$, which implies that f^* does not satisfy SID with any λ below this value. Note that the SID condition is used to guarantee that, for every cell $\mathcal{C}$, there exists a potential split that would decrease the residual variance within $\mathcal{C}$ by a constant factor of at most $1 - \lambda$. Combined with concentration of impurity decrease values, one can then form a recursive inequality of the form $J(l+1) \le (1 - \lambda/(1+\epsilon))J(l)$, where $J(l)$ denotes the squared bias of the CART tree grown to depth l and ϵ is a finite-sample error. Iterating gives $J(l) \le J(0)\exp(-\lambda l/(1+\epsilon))$, while the variance at depth l is $O(2^l/n)$, up to poly-log factors in n and d. Balancing bias and variance requires $\exp(-\lambda l/(1+\epsilon)) \approx 2^l/n$, that is, $l \approx \log n/(\lambda/(1+\epsilon) + \log 2)$, in which case we obtain the optimized risk bound $O(n^{-\lambda/((1+\epsilon)\log 2+\lambda)})$. Substituting $\lambda = (2^s - 1)^{-1}$ shows that the exponent $\lambda/((1+\epsilon)\log 2 + \lambda)$ is of the order 2^{-s}, so the bound is trivial unless $\log_2 n = \Omega(2^s)$. We will soon show that this is unnecessarily pessimistic, with the true sample complexity of the order $\log_2 n = 2s + o(1)$.

On the other hand, while the AND function has poor marginal signal strength at the root, the marginal signal strength improves as the tree gets deeper and more of the relevant covariates are selected. Indeed, we have $\mathrm{Corr}^2\{f^*(\mathbf{X}), X_j \mid X_1 = X_2 = \dots = X_{j-1} = 1\} = (2^{s-j} - 1)^{-1}$. By only quantifying the worst case signal strength, the SID condition is unable to capture and exploit this nuance.

To provide a sharper bound, we need to provide a new definition. We say that f^* has the λ-*stable merged staircase property*, denoted $f^* \in \mathsf{SMSP}(\lambda)$, if for any cell $\mathcal{C} \subset \{\pm 1\}^d$ such that $f|_{\mathcal{C}}$ is not constant, we have

$$\max_{k \in [d] \setminus J(\mathcal{C})} \mathrm{Cov}^2\{f^*(\mathbf{X}), X_k \mid \mathbf{X} \in \mathcal{C}\} 2^{-|J(\mathcal{C}) \cap S^*|} \ge \lambda. \tag{46}$$

Compared with the SID formula (7), this definition is stated in terms of the squared covariance and further modulates it with the base 2 exponential of the number of relevant covariates already split on. This quantity more accurately captures the signal-to-noise ratio for the problem of covariate selection within $\mathcal{C}$. Indeed, rather than attempt to control the amount of bias reduction at each split, we will instead lower bound the probability of selecting a relevant covariate. Formalizing this approach yields the following theorem.

THEOREM 7.5. *Under Assumption* 3.1, *let* $f_0^* \in \mathsf{SMSP}(\lambda)$. *Suppose* ε *is sub-Gaussian with parameter* σ [54]. *Denote*

$$\tau := 18\big(9\|f^*\|_\infty^2 + \sigma^2\big)\big((s+2)\log 3 + \log(2dn)\big) \tag{47}$$

and suppose the training sample size satisfies $n > \frac{8\tau}{\lambda}$. *Suppose further that we fit* $\hat{f}_{\mathrm{CART}}$ *with the stopping rule given by a minimum impurity decrease value* γ *satisfying* $\frac{\tau}{n} \le \gamma <$

$(\sqrt{\frac{\lambda}{2}} - \sqrt{\frac{\tau}{n}})_+^2$. *Such a value* γ *exists given the lower bound on* n. *We then have the expected* L^2 *risk upper bound*

$$\mathfrak{R}(\hat{f}_{\mathrm{G}}, f_0^*, d, n) = O\left(\frac{2^s(\|f^*\|_\infty^2 + \sigma^2)(s + \log n)}{n}\right). \tag{48}$$

In particular, $\hat{f}_{\mathrm{CART}}$ *is high-dimensional consistent for* f_0^*.

PROOF. See Section H of the Supplementary Material. □

Returning to the example of the AND function, it is easy to check that we may set λ in (46) to be 2^{-2s}. Plugging this into Theorem 7.5 shows that (48) holds whenever $n = \tilde{\Omega}(s2^{2s})$ in agreement with what was mentioned earlier.

8. Robust lower bounds for greedy trees. In this section, we will prove lower bounds for MSP functions that depend on how far they are from being non-MSP. Since being non-MSP is a somewhat fragile property and can be destroyed by small perturbations of the zero coefficients of the function, we call these robust lower bounds. In doing so, we will also strengthen the connection between the Fourier graph of a function f^* and its estimation properties using greedy trees and ensembles. These lower bounds will provide further performance gaps between greedy trees and ERM trees. Finally, comparing the lower bounds with the results from the previous section enable us to get a sense of how sharp both sets of bounds are.

To motivate the theorem and our proof technique, we first revisit the sth order AND function discussed in the previous section. We have already seen that $\tilde{O}(s2^{2s})$ samples are sufficient for CART to achieve small estimation error. While a much smaller number than what was previously achievable using the SID assumption, it is still larger than the upper bound of $\tilde{O}(2^s)$ achievable using ERM trees (Proposition 6.2). Nonetheless, the weak upper bound turns out to be relatively sharp. To see this, we write

$$\mathbb{1}\{x_j = 1 \colon j = 1, 2, \ldots, s\} = \prod_{j=1}^{s}\left(\frac{x_j + 1}{2}\right) = 2^{-s} + \sum_{j=1}^{s} 2^{-s}x_j + \text{higher-order terms}. \tag{49}$$

From this representation, we see that all first-order terms have small coefficients when s is relatively large, and if removed, will turn the function into a non-MSP one.

This argument can be made rigorous via the technique of coupling. Let f^* denote the AND function and let define $\check{f}^*$ via $\check{f}^*(\mathbf{x}) = f^*(\mathbf{x}) - \sum_{j=1}^{s} 2^{-s}x_j$. Now define a dataset $\check{\mathcal{D}}_n := \{(\mathbf{X}_i, \check{Y}_i)\}_{i=1}^n$ with $\check{Y}_i = \check{f}^*(\mathbf{X}_i) + \varepsilon_i$ for $i = 1, 2, \ldots, n$. If $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n \sim \mathcal{N}(0, \sigma^2)$, it is easy to check that the Kullback–Leibler divergence between the original and modified datasets satisfies

$$\mathrm{KL}(\mathcal{D}_n \| \check{\mathcal{D}}_n) = \frac{s2^{-2s}n}{2\sigma^2}. \tag{50}$$

Using Pinsker's inequality, this gives a bound on the total variation distance, which we use to argue that there is a coupling between the two datasets with a probability at least $1 - \sqrt{\frac{s2^{-2s}n}{4\sigma^2}}$ event on which they are equal. The probability of this event is thus at least $1/2$ unless $n = \Omega(2^{2s}/s)$, and on this event, a lower bound on the estimation error for $\check{f}^*$ translates to one on the estimation error for f^*.

This coupling technique can be extended to handle other MSP functions. Considering the Fourier graph of the AND function, we see that the vertices $\{1\}, \{2\}, \ldots, \{s\}$ form a *vertex cut* whose removal disconnects the vertex $\varnothing$ from the rest of the graph. More generally, if we can identify a vertex cut $\mathcal{B}$ for the Fourier graph of some function $f^* = \sum_{S \in \mathcal{S}} \alpha_S \chi_S$, its weight $w(\mathcal{B})$ quantifies the L^2 distance between f^* and the non-MSP function $f^*_{-\mathcal{B}} := \sum_{S \in \mathcal{S} \setminus \mathcal{B}} \alpha_S \chi_S$. Any estimation error lower bound for $f^*_{-\mathcal{B}}$ can therefore be translated into a lower bound for f^* on the probability at least $1 - \sqrt{\frac{w(\mathcal{B})n}{2\sigma^2}}$ event that datasets generated with these regression functions can be coupled.

To state this more formally, we first note that the Fourier graph of $f^*_{-\mathcal{B}}$ is equivalent to the induced subgraph of $\mathcal{S} \setminus \mathcal{B}$ in $\mathcal{G}_{f^*}$, denoted $\mathcal{G}_{f^*}[\mathcal{S} \setminus \mathcal{B}]$. Denote the connected component of the empty set $\varnothing$ in $\mathcal{G}_{f^*}[\mathcal{S} \setminus \mathcal{B}]$ as $\mathcal{S}_{-\mathcal{B},\mathsf{MSP}}$ and set $S^*_{-\mathcal{B},\mathsf{MSP}} := \bigcup_{S \in \mathcal{S}_{-\mathcal{B},\mathsf{MSP}}} S$. Furthermore, let $\mathcal{S}_{-\mathcal{B},-\mathsf{MSP}} := \mathcal{S} \setminus (\mathcal{S}_{-\mathcal{B},\mathsf{MSP}} \cup \mathcal{B})$ denote the subcollection comprising all vertices disconnected from $\varnothing$ in $\mathcal{G}_{f^*}[\mathcal{S} \setminus \mathcal{B}]$. We have the following elaboration on Theorem 5.1.

THEOREM 8.1. *Under Assumption* 3.1, *suppose* $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n \sim_{i.i.d.} \mathcal{N}(0, \sigma^2)$, *and let* $\mathcal{G}_{f_0^*}$ *be the Fourier graph of* f_0^* *with weight function* w. *Let* $\mathcal{B} \subset \mathcal{S}$ *be a subset of vertices of* $\mathcal{G}_{f_0^*}$. *Let* s_{T} *be the minimum size of a traversal* T *for* $\mathcal{S}_{-\mathcal{B},-\mathsf{MSP}}$ *such that* $T \subset [d] \setminus S^*_{-\mathcal{B},\mathsf{MSP}}$ *and let* $s_{-\mathcal{B},\mathsf{MSP}} := |S^*_{-\mathcal{B},\mathsf{MSP}}|$. *For any* $0 < \delta < 1$, *suppose*

$$\log_2 n \le \min\left\{\frac{\delta(d - s_{-\mathcal{B},\mathsf{MSP}} - s_{\mathrm{T}})}{2 s_{\mathrm{T}}} - 2,\ \log_2 \frac{\delta^2 \sigma^2}{w(\mathcal{B})} - 1\right\}. \tag{51}$$

Then the expected L^2 *risk of any greedy tree model satisfies*

$$\mathfrak{R}(\hat{f}_{\mathrm{G}}, f_0^*, d, n) \ge (1 - \delta) w(\mathcal{S}_{-\mathcal{B},-\mathsf{MSP}}). \tag{52}$$

PROOF. See Section I of the Supplementary Material. □

As seen from the AND function, the sample complexity bound in Theorem 8.1 can be relatively sharp. This is not always the case however. Consider, for instance, $f^* = \sum_{j=1}^{s-1} \chi_{\{1,2,\ldots,j\}} + \alpha \chi_{\{1,2,\ldots,s\}} + \sum_{j=s+1}^{2s} \chi_{\{1,2,\ldots,j\}}$. Applying Theorem 8.1 to this function gives a sample complexity lower bound of $\min\{\Omega(1/\alpha^2), \exp(\Omega(d))\}$. On the other hand, we have

$$\max_{k \in \{s,s+1,\ldots,d\}} \mathrm{Cov}^2\{f^*(\mathbf{X}), X_k \mid X_1, X_2, \ldots, X_{s-1}\} 2^{-s} = \alpha^2 2^{-s}, \tag{53}$$

which implies that the sample complexity upper bound from Theorem 7.5 is $\Omega(2^s/\alpha^2)$. One may show that this upper bound is sharp—due to the structure of f^*, with enough samples, CART will only attempt to split on x_s at depth s, in which case the sample size available for determining this split is approximately $n/2^s$. This nuance is not captured by our lower bound, yet any attempt to address it seems to require sacrificing some aspect of the generality of our result. We hence leave this effort to future work.

REMARK 8.2. Since (39), the lower bounds for greedy tree ensembles, requires a bounded noise assumption, it does not immediately extend to a robust lower bound, which requires a Gaussian noise assumption. This can be overcome by conditioning on the highly probable event on which $\max_{1 \le i \le n} |Y_i| \le C\sigma\sqrt{\log n}$ for some universal constant C.

9. Extensions of lower bounds. The results in our paper have been presented in the setting of binary covariates under the uniform distribution. Nonetheless, our lower bound proof techniques also apply more generally, which we illustrate in this section. First, recall that our strategy has been to (i) derive an upper bound on the probability of selecting relevant covariates (cf., Lemma 4.2) and then (ii) turn this into a lower bound on estimation error (cf., Lemma 3.2 and Lemma 3.3). The arguments for the second step are very general and depend only on the independence between the relevant covariates and the irrelevant covariates. As such, aside from this consideration, the barrier to any extension lies with the the first step. With this in mind, we illustrate two extensions of the bounds to continuous covariates. That is, we study the setting $\mathcal{X} = [0, 1]^d$, $\nu = \mathrm{Unif}([0, 1]^d)$.

Note also that in the continuous setting, CART and RFs are invariant to rescaling of the covariate values, such that any analysis of the uniform distribution immediately extends to distributions with continuous densities and independent covariates. More precisely, consider product measures $\nu = \bigotimes_{j=1}^d \nu_j$ on $\mathbb{R}^d$ whose marginal distributions ν_j admit continuous c.d.f.s (not necessarily with compact support). Applying the probability integral transform componentwise maps such a ν to the uniform distribution on $[0, 1]^d$. Since CART and RFs depend only on the ordering of feature values, this coordinatewise monotone reparametrization leaves the fitted trees, and hence our lower bounds, unchanged.

EXAMPLE 9.1. Consider the original CART and RF algorithms [26]. The XOR function can be extended to continuous covariates as a step function:

$$f^*(\mathbf{x}) = \mathbb{1}\{x_1 x_2 \geq 1/2\} - \mathbb{1}\{x_1 x_2 < 1/2\}. \tag{54}$$

Similar to discrete covariate setting, there is marginal signal bottleneck, and we may almost completely imitate the proof of Lemma 4.2 to obtain an estimation error lower bound for the CART and RF estimators whenever $\log n \leq Cd$ for some universal C. The only caveat is that we can no longer use the results from Section C of the Supplementary Material to bound the query path length to be $O(\log n)$. To regain such a bound, it suffices to ensure relatively balanced splits on irrelevant covariates with high probability, which can be enforced in practice by restricting, at each split on a continuous covariate within a cell $\mathcal{C}$, the candidate thresholds to lie between the empirical α- and $(1 - \alpha)$-quantiles (for some fixed $\alpha \in (0, 1/2)$) of that covariate in $\mathcal{C}$. Under the assumption of independent covariates with continuous densities bounded away from 0 and ∞, this implies that both child cells $\mathcal{C}_L$ and $\mathcal{C}_R$ have Lebesgue measure that is a constant fraction smaller than their parent.

Observe that the more general lower bounds in Theorem 5.1 and Theorem 8.1 also generalize, with the modification that each monomial χ_S is redefined as

$$\chi_S(\mathbf{x}) := \prod_{j \in S} 2^{-1/2}\big(\mathbb{1}\{x_j \geq 1/2\} - \mathbb{1}\{x_j < 1/2\}\big). \tag{55}$$

However, unlike the binary covariate setting in which these monomials form a basis for the entire space of L^2 functions, these are but a small subset of $L^2([0, 1]^d)$, which makes the theory less satisfying.

On the other hand, we note that $L^2([0, 1]^d)$ possesses a wavelet basis, which is constructed as follows: For each depth $l = 0, 1, 2, \ldots$, for $m = 1, 2, \ldots, 2^l$, define the univariate function

$$\psi_{l,m}(x) := -2^{l/2}\mathbb{1}\left\{\frac{m-1}{2^l} < x \leq \frac{m-1/2}{2^l}\right\} + 2^{l/2}\mathbb{1}\left\{\frac{m-1/2}{2^l} < x \leq \frac{m}{2^l}\right\}. \tag{56}$$

As we vary l and m, this forms the Haar wavelet basis for $L^2([0, 1])$, and we can extend it to a basis for $L^2([0, 1]^d)$ by taking products. Just as the Fourier basis gives a convenient basis for analyzing CART when applied to binary covariates, the product Haar wavelet basis is convenient for analyzing dyadic CART, whose splits are constrained to be at the midpoint of each side of a given cell. Indeed, the impurity decrease for splitting on a covariate X_k in a cell $\mathcal{C}$ can be written as $\widehat{\mathrm{Cov}}^2\{Y, \psi_{l,m}(X_k) \mid \mathbf{X} \in \mathcal{C}\}$ for an appropriate choice of (l, m).

EXAMPLE 9.2. Consider the sine function $f^*(\mathbf{x}) = \sin(\pi x_1)$. One can show that for any cell $\mathcal{C}$, unless $\mathcal{C}$ has already split on x_1, $f^*(\mathbf{X})$ is conditionally independent of $\psi_{0,1}(X_1)$ in $\mathcal{C}$. Using a modification of our coupling technique, one can therefore show that dyadic CART, when fitted to data generated from this function, will make random splits. We therefore obtain a constant lower bound for the estimation error of dyadic CART and RF estimators whenever $\log_2 n \le Cd$ for some universal C.

To see how far we can push this result, we next construct the equivalent of the Fourier graph. To this end, we first index each basis element using a collection of tuples, that is, given $\Lambda := \{(k_j, l_j, m_j) : j = 1, 2, \ldots, r\}$, define $\chi_\Lambda(\mathbf{x}) := \prod_{j=1}^r \psi_{l_j,m_j}(x_{k_j})$. Now given any function $f^* \in L^2([0,1]^d)$, we may write it as $f^* := \sum_{\Lambda\in\mathfrak{L}} \alpha_\Lambda \chi_\lambda$. Define the directed hypergraph $\mathcal{G}$ with vertex set $\mathfrak{L}$ and put an edge $\{\Lambda^{(1)}, \Lambda^{(2)}, \ldots, \Lambda^{(q)}\} \to \Lambda^{(q+1)}$ if the following hold: (i) the union $\bigcup_{j=1}^q \Lambda^{(j)}$ contains at most one entry for each covariate, (ii) $|\Lambda^{(q+1)} \setminus \bigcup_{j=1}^q \Lambda^{(j)}| \le 1$ and (iii) if $\Lambda^{(q+1)} \setminus \bigcup_{j=1}^q \Lambda^{(j)} = \{(k,l,m)\}$, then $(k, l-1, \lceil m/2 \rceil) \in \bigcup_{j=1}^q \Lambda^{(j)}$. Similar to the binary case, we define a weight function via $w(\Lambda) := \alpha_\Lambda^2$ and call $(\mathcal{G}, w)$ the *wavelet graph* of f^*.

The natural analogue of MSP is connectedness in the wavelet graph and one would expect the absence of this condition to imply high-dimensional inconsistency. Unfortunately, disconnectedness alone does not imply the conditional independence we need for our previous proofs to go through. For instance, if we set $f^*(\mathbf{x}) = \psi_{1,1}(x_1)$, then $\{(1,1,1)\}$ is disconnected from $\varnothing$ in the wavelet graph. While this implies that the random variable $\psi_{0,1}(X_1)$, which corresponds to a first split on x_1, is uncorrelated with the response variable Y in any cell, the two variables are still clearly dependent. We leave how to address this issue to future work.

Finally, we note that while our current analysis is tailored to the uniform distribution and relies on the structure of Fourier or wavelet bases, recent work in computational learning theory has pursued more distribution-agnostic complexity characterizations. For example, [29] analyze the statistical and gradient query complexity of learning sparse functions under general product measures.

10. Overcoming marginal signal bottleneck. The proof of our lower bounds isolates a source of the potential poor performance of greedy trees: It stems from their use of marginal projections of the dataset (cf., equation (17)) to determine splits instead of using its joint distribution. While this increases computational efficiency, it runs into problems when large regression function heterogeneity is masked by weak marginal signal. We called this problem *marginal signal bottleneck*. In this section, we discuss the ability of other classes of regression tree algorithms to avoid the marginal signal bottleneck problem and the computation price each incurs in doing so.

Within the family of greedy algorithms, [36]'s GUIDE algorithm tests for pairwise interactions when selecting splits and is thus able to estimate the XOR function. On the other hand, it is unable to estimate higher-order monomials. One can generalize this idea in a binary feature setting to permit splits on k-way monomials $\prod_{j\in S} x_j$ with $|S| \le k$. Such an algorithm can indeed learn leap-k regression functions (e.g., order k monomials, see also Section K of the Supplementary Material, under a genericity condition on Fourier coefficients, with a much lower sample complexity of $O(k \log d)$, but at a much higher computational cost of $O(nd^k)$ per split. However, for functions beyond leap-k (e.g., monomials of order larger than k), all conditional correlations of degree $\le k$ vanish, so consistency still requires $\exp(\Omega(d))$ samples. Alternatively, CART can be modified to allow for splits that are oblique, that is, along linear hyperplanes $\sum_{j=1}^d \beta_j x_j$ depending on multiple covariates [15]. This creates fewer opportunities for signal bottleneck—for instance, the XOR function in continuous covariates is no longer an issue. However, optimizing over oblique splits is vastly more computationally intensive and also reduces the interpretability of the resulting model.

Bayesian CART [19] and its ensemble version, Bayesian Additive Regression Trees (BART) [20] put regression trees in a Bayesian framework and use a Metropolis–Hastings strategy to sample from its posterior. Each MCMC iteration can be thought of as a stochastic update of the tree structure and can potentially escape local minima in which deterministic methods may get stuck. However, [48] was able to show that the sampler mixes slowly and in fact becomes increasingly greedy as the training data increases, leading to problems with estimating the XOR function.

Due to improvements in hardware as well as generic optimization solvers, directly solving the ERM problem for decision trees has become much more feasible than before and is currently an active area of research. Various approaches have been formulated, including mixed integer linear programming [7, 11, 52, 59] and dynamic programming [8, 21, 27, 35, 58]. Most of these works focus solely on algorithms for classification trees, with [58] the only work we are aware of that as introduced an algorithm for optimal regression trees. On the theoretical front, it is interesting to examine if this class of algorithms can provably overcome the limitations of greedy trees and their ensembles, under polynomial runtime constraints.

11. Simulations. To supplement our theory, we perform a simulation study in which we generate training data according to (1), with the covariates distributed uniformly on $\{\pm 1\}^d$ and the regression function chosen to be $f^*(\mathbf{x}) = x_1x_2 + \alpha x_1$. This corresponds to the XOR function when $\alpha = 0$, while setting nonzero α allows for some amount of marginal signal. We vary $d \in \{10, 20, 30, 40, 50\}$, $\log_2 n \in \{7, 9, 11, 13, 15\}$, $\alpha \in \{0, 0.02\}$, and $\sigma^2 \in \{0, 0.1\}$. We use two metrics to evaluate the performance of CART on each dataset:

- Expected mean squared error (MSE) $R(\hat{f}_{\mathrm{CART}}, f^*)$, where $\hat{f}_{\mathrm{CART}}$ is fitted with a minimum impurity decrease stopping criterion (14). Here, the threshold γ is selected based on a held-out validation set.
- Split coverage of each feature X_k by a CART tree $\mathfrak{T}$ that is grown to purity, that is, with no stopping criterion. Split coverage is defined as

$$\text{coverage}(k, \mathfrak{T}) := \sum_{\mathcal{L}\in\mathcal{P}(\mathfrak{T})} 2^{-\text{depth}(\mathcal{L})}\mathbb{1}\{\text{path from root to } \mathcal{L} \text{ splits on } X_k\}. \tag{57}$$

  For the uniform measure, coverage$(k, \mathfrak{T})$ computes the probability that the root-to-leaf path in $\mathfrak{T}$ followed by a random query point $\mathbf{X}$ splits on the feature X_k.

Selected results from the experiment, averaged over 200 replicates, are shown in Figure 1 and Figure 2. In each panel of Figure 1, we plot how expected MSE varies across the grid of n and d values, stratifying on all other hyperparameters of the data generating process. In both panels, we focus on the noiseless setting ($\sigma^2 = 0$), under which the null model satisfies $R(0, f^*) = 1 + \alpha^2$. The left panel corresponds to the XOR setting ($\alpha = 0$), while the right panel corresponds to $\alpha = 0.02$. Note that while d varies on a linear scale, n varies on a logarithmic scale. Nonetheless, in the left panel, we see that the expected MSE values increase along the bottom-left to top-right diagonal, reflecting that the sample complexity required to "learn" the regression function has exponential dependence on the ambient dimension, which is in agreement with Proposition 4.1. This exponential dependence disappears in the right panel, even though the energy of the additional term, $0.02X_1$, is much smaller than that of the main term X_1X_2. This agrees with the high-dimensional consistency guarantee provided in Theorem 7.5. Figure 2 similarly plots how split coverage varies across the grid of n and d values, fixing $\alpha = 0$ and $\sigma^2 = 0$. For this figure, however, the left panel corresponds to the coverage of X_2, a relevant feature, while the right panel corresponds to the

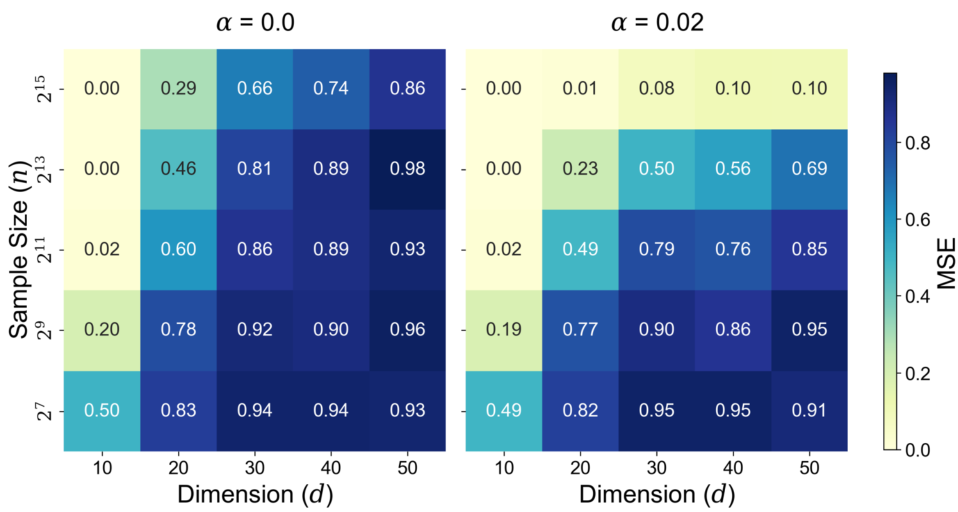

FIG. 1. *Expected MSE $R(\hat{f}_{\mathrm{CART}}, f)$ for CART fitted to $Y = X_1X_2 + \alpha X_1$, with $\alpha = 0$ (left panel) and $\alpha = 0.02$ (right panel). CART is fitted with a minimum impurity decrease stopping criterion. We observe high-dimensional consistency in the right panel since the true function satisfies* MSP*and lack of high-dimensional consistency in the left panel since the true function does not satisfy* MSP.

coverage of X_3, which is an irrelevant feature. In the left panel, we see that coverage of X_2 has a decreasing trend across the bottom-left to top-right diagonal, which again shows that the sample complexity required for reliably performing feature selection grows exponentially in d, in agreement with Lemma 4.2. The right panel gives reference values for coverage of an irrelevant feature. One may observe that these values are upper bounded by $\log_2 n/d$. Additional simulation results, including those for $\sigma^2 = 0.01$, are shown in Section L of the Supplementary Material.

12. Discussion. In this paper, we have provided rigorous estimation error lower bounds for CART and RFs that show they have the potential to get stuck in local minima. In fact, we have been able to go much further by evaluating the performance of regression tree algorithms

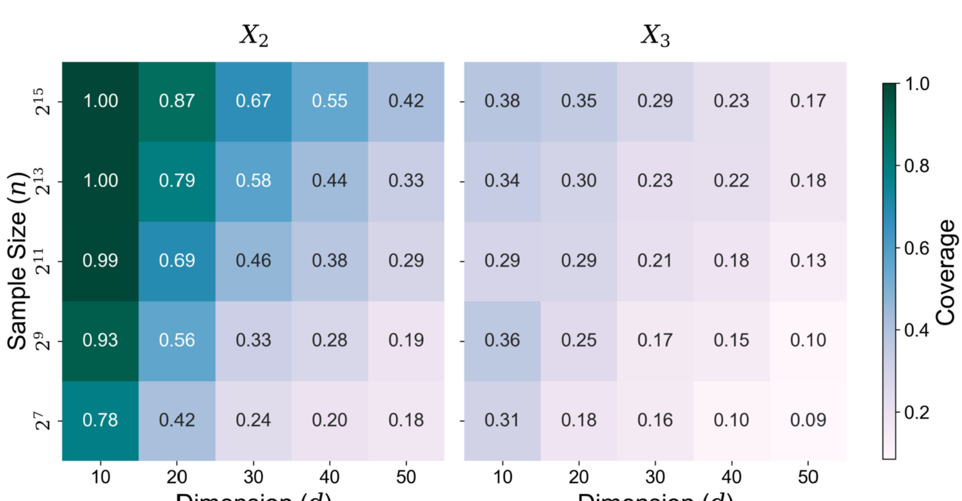

FIG. 2. *Split coverage for features X_2 and X_3 for CART fitted to $Y = X_1X_2$, without any stopping criterion. Split coverage is defined in (57). We observe that the sample size required to reliably split on the relevant feature grows exponentially in d.*

in terms of high-dimensional consistency. We have used the MSP to completely characterize the regression functions for which CART is high-dimensional consistent and secondly established performance gaps between greedy and globally optimal tree algorithms over a large class of regression functions.

Our work adds to the small but growing literature on negative results for regression trees and ensembles. [50] showed that RFs can be inconsistent when its constituent trees are not diverse enough. [46] proved that individual regression trees are inefficient at estimating additive structure, regardless of the way in which they are optimized. [16] demonstrate that when trees are fitted with CART to a constant signal model, they tend to place splits near the boundaries of the covariate space. This tendency can result in nodes containing very few samples—a significant problem when uniform control over estimation accuracy is desired, such as in the context of heterogeneous treatment effect estimation. Finally, [48] proved that mixing times for Bayesian additive regression trees can increase with the training sample size.

These results, especially when used as part of a head-to-head comparison against other methods, sound a note of caution that regression trees and ensembles should not be used blindly, despite their popularity—in certain situations, their output can be unexpectedly inaccurate or vastly inferior to that of other methods. They also suggest ways in which these methods can be modified to overcome their shortcomings or justify changes that have already been proposed.

Finally, while we have proved lower bounds for greedy axis-aligned regression trees, it is natural to ask whether lower bounds of a similar flavor hold for oblique regression trees as well as other more general classes of regression trees and ensembles under computational constraints. We hope our work will serve as a first step in clarifying the statistical-computational trade-offs inherent within the class of greedy algorithms fitting regression trees and ensembles.

Acknowledgments. The authors would like to thank Yingying Fan, Samory Kpotufe, Wei-Yin Loh, Rahul Mazumder, Denny Wu and Qiong Zhang for insightful discussions.

Funding. Tan gratefully acknowledges support from NUS Start-up Grant A-8000448-00-00 and MOE AcRF Tier 1 Grant A-8002498-00-00.

Klusowski gratefully acknowledges financial support from the National Science Foundation (NSF) through CAREER DMS-2239448.

Balasubramanian gratefully acknowledges support from NSF through grants DMS-2053918 and DMS-2413426.

SUPPLEMENTARY MATERIAL

Supplement to "Statistical-computational trade-offs for recursive adaptive partitioning estimators." (DOI: 10.1214/25-AOS2603SUPP; .pdf). Section A discusses related work on consistency of ERM trees, greedy trees and nonadaptive trees. Section B derives a lower bound for nonadaptive trees. Section C derives an upper bound for the length of query paths. Section D provides a proof of Theorem 5.1. Section E provides a proof of Proposition 6.1. Section F provides a proof of Proposition 6.2. Section G provides some results regarding the stable MSP property. Section H provides a proof of Theorem 7.5. Section I provides a proof of Theorem 8.1. Section J provides constant probability lower bounds. Section K makes comparisons between greedy regression trees and neural networks trained by SGD. Section L provides additional simulation results.

REFERENCES

[1] ABBE, E., ADSERA, E. B. and MISIAKIEWICZ, T. (2022). The merged-staircase property: A necessary and nearly sufficient condition for SGD learning of sparse functions on two-layer neural networks. In *Conference on Learning Theory* 4782–4887. PMLR.

[2] ABBE, E., ADSERA, E. B. and MISIAKIEWICZ, T. (2023). SGD learning on neural networks: Leap complexity and saddle-to-saddle dynamics. In *The Thirty Sixth Annual Conference on Learning Theory* 2552–2623. PMLR.

[3] ABBE, E. and BOIX-ADSERÀ, E. (2022). On the non-universality of deep learning: Quantifying the cost of symmetry. In *Advances in Neural Information Processing Systems* (A. H. Oh, A. Agarwal, D. Belgrave and K. Cho, eds.).

[4] ABBE, E., BOIX-ADSERA, E., BRENNAN, M., BRESLER, G. and NAGARAJ, D. (2021). The staircase property: How hierarchical structure can guide deep learning.

[5] ABBE, E. and SANDON, C. (2023). Polynomial-time universality and limitations of deep learning. *Comm. Pure Appl. Math.* **76** 3493–3549. MR4642823 https://doi.org/10.1002/cpa.22121

[6] AGARWAL, A., TAN, Y. S., RONEN, O., SINGH, C. and YU, B. (2022). Hierarchical shrinkage: Improving the accuracy and interpretability of tree-based models. In *International Conference on Machine Learning* 111–135. PMLR.

[7] AGHAEI, S., GÓMEZ, A. and VAYANOS, P. (2025). Strong optimal classification trees. *Oper. Res.* **73** 2223–2241. MR4958675

[8] AGLIN, G., NIJSSEN, S. and SCHAUS, P. (2020). Learning optimal decision trees using caching branch-and-bound search. *Proc. AAAI Conf. Artif. Intell.* **34** 3146–3153.

[9] ATHEY, S. and IMBENS, G. (2016). Recursive partitioning for heterogeneous causal effects. *Proc. Natl. Acad. Sci. USA* **113** 7353–7360. MR3531135 https://doi.org/10.1073/pnas.1510489113

[10] BARAK, B., EDELMAN, B., GOEL, S., KAKADE, S., MALACH, E. and ZHANG, C. (2022). Hidden progress in deep learning: SGD learns parities near the computational limit. *Adv. Neural Inf. Process. Syst.* **35** 21750–21764.

[11] BERTSIMAS, D. and DUNN, J. (2017). Optimal classification trees. *Mach. Learn.* **106** 1039–1082. MR3665788 https://doi.org/10.1007/s10994-017-5633-9

[12] BREIMAN, L. (2001). Random forests. *Mach. Learn.* **45** 5–32.

[13] BREIMAN, L., FRIEDMAN, J. H., OLSHEN, R. A. and STONE, C. J. (1984). *Classification and Regression Trees*. *Wadsworth Statistics/Probability Series*. Wadsworth Advanced Books and Software, Belmont, CA. MR0726392

[14] CARUANA, R., KARAMPATZIAKIS, N. and YESSENALINA, A. (2008). An empirical evaluation of supervised learning in high dimensions. In *Proceedings of the* 25*th International Conference on Machine Learning* 96–103.

[15] CATTANEO, M. D., CHANDAK, R. and KLUSOWSKI, J. M. (2024). Convergence rates of oblique regression trees for flexible function libraries. *Ann. Statist.* **52** 466–490. MR4744184 https://doi.org/10.1214/24-aos2354

[16] CATTANEO, M. D., KLUSOWSKI, J. M. and TIAN, P. M. (2022). On the pointwise behavior of recursive partitioning and its implications for heterogeneous causal effect estimation. Preprint. Available at arXiv:2211.10805.

[17] CHEN, T. and GUESTRIN, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the* 22*nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 785–794.

[18] CHI, C.-M., VOSSLER, P., FAN, Y. and LV, J. (2022). Asymptotic properties of high-dimensional random forests. *Ann. Statist.* **50** 3415–3438. MR4524502 https://doi.org/10.1214/22-aos2234

[19] CHIPMAN, H. A., GEORGE, E. I. and MCCULLOCH, R. E. (1998). Bayesian CART model search. *J. Amer. Statist. Assoc.* **93** 935–948.

[20] CHIPMAN, H. A., GEORGE, E. I. and MCCULLOCH, R. E. (2010). BART: Bayesian additive regression trees. *Ann. Appl. Stat.* **4** 266–298. MR2758172 https://doi.org/10.1214/09-AOAS285

[21] DEMIROVIĆ, E., LUKINA, A., HEBRARD, E., CHAN, J., BAILEY, J., LECKIE, C., RAMAMOHANARAO, K. and STUCKEY, P. J. (2022). MurTree: Optimal decision trees via dynamic programming and search. *J. Mach. Learn. Res.* **23** Paper No. [26], 47. MR4420751

[22] FERNÁNDEZ-DELGADO, M., CERNADAS, E., BARRO, S. and AMORIM, D. (2014). Do we need hundreds of classifiers to solve real world classification problems? *J. Mach. Learn. Res.* **15** 3133–3181. MR3277155

[23] FRIEDMAN, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Ann. Statist.* **29** 1189–1232. MR1873328 https://doi.org/10.1214/aos/1013203451

[24] GLASGOW, M. R. (2024). SGD finds then tunes features in two-layer neural networks with near-optimal sample complexity: A case study in the XOR problem. In *The Twelfth International Conference on Learning Representations*.

[25] GRINSZTAJN, L., OYALLON, E. and VAROQUAUX, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Adv. Neural Inf. Process. Syst.* **35** 507–520.

[26] HASTIE, T., TIBSHIRANI, R. and FRIEDMAN, J. (2001). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer Series in Statistics*. Springer, New York. MR1851606 https://doi.org/10.1007/978-0-387-21606-5

[27] HU, X., RUDIN, C. and SELTZER, M. (2019). Optimal sparse decision trees. *Adv. Neural Inf. Process. Syst.*.

[28] ISHWARAN, H., KOGALUR, U. B., BLACKSTONE, E. H. and LAUER, M. S. (2008). Random survival forests. *Ann. Appl. Stat.* **2** 841–860. MR2516796 https://doi.org/10.1214/08-AOAS169

[29] JOSHI, N., MISIAKIEWICZ, T. and SREBRO, N. (2024). On the complexity of learning sparse functions with statistical and gradient queries. In *The Thirty-Eighth Annual Conference on Neural Information Processing Systems*.

[30] KE, G., MENG, Q., FINLEY, T., WANG, T., CHEN, W., MA, W., YE, Q. and LIU, T.-Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* **30**.

[31] KLUSOWSKI, J. M. (2020). Sparse learning with CART. In *Advances in Neural Information Processing Systems* (H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan and H. Lin, eds.) **33** 11612–11622. Curran Associates, Red Hook.

[32] KLUSOWSKI, J. M. and TIAN, P. (2021). Nonparametric variable screening with optimal decision stumps. In *Proceedings of the* 24*th International Conference on Artificial Intelligence and Statistics* (A. Banerjee and K. Fukumizu, eds.). *Proceedings of Machine Learning Research* **130** 748–756. PMLR.

[33] KLUSOWSKI, J. M. and TIAN, P. M. (2024). Large scale prediction with decision trees. *J. Amer. Statist. Assoc.* **119** 525–537. MR4713911 https://doi.org/10.1080/01621459.2022.2126782

[34] KOU, Y., CHEN, Z., GU, Q. and KAKADE, S. M. (2024). Matching the statistical query lower bound for k-sparse parity problems with stochastic gradient descent. Preprint. Available at arXiv:2404.12376.

[35] LIN, J., ZHONG, C., HU, D., RUDIN, C. and SELTZER, M. (2020). Generalized and scalable optimal sparse decision trees. In *International Conference on Machine Learning* 6150–6160. PMLR.

[36] LOH, W.-Y. (2002). Regression trees with unbiased variable selection and interaction detection. *Statist. Sinica* **12** 361–386. MR1902715

[37] MAZUMDER, R. and WANG, H. (2024). On the convergence of CART under sufficient impurity decrease condition. *Adv. Neural Inf. Process. Syst.* **36**.

[38] MEI, S., MONTANARI, A. and NGUYEN, P.-M. (2018). A mean field view of the landscape of two-layer neural networks. *Proc. Natl. Acad. Sci. USA* **115** E7665–E7671. MR3845070 https://doi.org/10.1073/pnas.1806579115

[39] MENDELSON, S. (2018). Learning without concentration for general loss functions. *Probab. Theory Related Fields* **171** 459–502. MR3800838 https://doi.org/10.1007/s00440-017-0784-y

[40] MURDOCH, W. J., SINGH, C., KUMBIER, K., ABBASI-ASL, R. and YU, B. (2019). Definitions, methods, and applications in interpretable machine learning. *Proc. Natl. Acad. Sci. USA* **116** 22071–22080. MR4030584 https://doi.org/10.1073/pnas.1900654116

[41] OLSON, R. S., CAVA, W. L., MUSTAHSAN, Z., VARIK, A. and MOORE, J. H. (2018). Data-driven advice for applying machine learning to bioinformatics problems. In *Biocomputing* 2018: *Proceedings of the Pacific Symposium* 192–203. World Scientific, Singapore.

[42] PEDREGOSA, F., VAROQUAUX, G., GRAMFORT, A. et al. (2011). Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.* **12** 2825–2830. MR2854348

[43] RUDIN, C., CHEN, C., CHEN, Z., HUANG, H., SEMENOVA, L. and ZHONG, C. (2022). Interpretable machine learning: Fundamental principles and 10 grand challenges. *Stat. Surv.* **16** 1–85. MR4361744 https://doi.org/10.1214/21-ss133

[44] STEIN, E. M. and SHAKARCHI, R. (2003). *Fourier Analysis: An Introduction. Princeton Lectures in Analysis* **1**. Princeton Univ. Press, Princeton, NJ. MR1970295

[45] SYRGKANIS, V. and ZAMPETAKIS, M. (2020). Estimation and inference with trees and forests in high dimensions. In *Conference on Learning Theory* 3453–3454. PMLR.

[46] TAN, Y. S., AGARWAL, A. and YU, B. (2022). A cautionary tale on fitting decision trees to data from additive models: Generalization lower bounds. In *International Conference on Artificial Intelligence and Statistics* 9663-9685. PMLR.

[47] TAN, Y. S, KLUSOWSKI, J. M and BALASUBRAMANIAN, K. (2026). Supplement to "Statistical-computational trade-offs for recursive adaptive partitioning estimators." https://doi.org/10.1214/25-AOS2603SUPP

[48] TAN, Y. S., RONEN, O., SAARINEN, T. and YU, B. (2024). The computational curse of big data for Bayesian Additive Regression Trees: a hitting time analysis. Preprint. Available at arXiv:2406.19958.

[49] TAN, Y. S. and VERSHYNIN, R. (2023). Online stochastic gradient descent with arbitrary initialization solves non-smooth, non-convex phase retrieval. *J. Mach. Learn. Res.* **24** Paper No. [58], 47. MR4582480

[50] TANG, C., GARREAU, D. and VON LUXBURG, U. (2018). When do random forests fail? In *Advances in Neural Information Processing Systems* **31** (S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi and R. Garnett, eds.). Curran Associates, Red Hook.

[51] UHLER, C., RASKUTTI, G., BÜHLMANN, P. and YU, B. (2013). Geometry of the faithfulness assumption in causal inference. *Ann. Statist.* **41** 436–463. MR3099109 https://doi.org/10.1214/12-AOS1080

[52] VERWER, S. and ZHANG, Y. (2019). Learning optimal classification trees using a binary linear program formulation. *Proc. AAAI Conf. Artif. Intell.* **33** 1625–1632.

[53] WAHBA, G. (1990). *Spline Models for Observational Data*. *CBMS-NSF Regional Conference Series in Applied Mathematics* **59**. SIAM, Philadelphia, PA. MR1045442 https://doi.org/10.1137/1.9781611970128

[54] WAINWRIGHT, M. J. (2019). *High-Dimensional Statistics*: *A Non-asymptotic Viewpoint*. *Cambridge Series in Statistical and Probabilistic Mathematics* **48**. Cambridge Univ. Press, Cambridge. MR3967104 https://doi.org/10.1017/9781108627771

[55] WASSERMAN, L. and LAFFERTY, J. (2005). Rodeo: Sparse nonparametric regression in high dimensions. *Adv. Neural Inf. Process. Syst.* **18**.

[56] WRIGHT, M. N., ZIEGLER, A. et al. (2017). ranger: A fast implementation of random forests for high dimensional data in C++ and R. *J. Stat. Softw.* **77**.

[57] WU, C. F. J. and HAMADA, M. S. (2009). *Experiments*: *Planning*, *Analysis*, *and Optimization*, 2nd ed. *Wiley Series in Probability and Statistics*. Wiley, Hoboken, NJ. MR2583259

[58] ZHANG, R., XIN, R., SELTZER, M. and RUDIN, C. (2023). Optimal sparse regression trees. *Proc. AAAI Conf. Artif. Intell.* **37** 11270–11279.

[59] ZHU, H., MURALI, P., PHAN, D., NGUYEN, L. and KALAGNANAM, J. (2020). A scalable MIP-based method for learning optimal multivariate decision trees. *Adv. Neural Inf. Process. Syst.* **33** 1771–1781.